Linear Stability Analysis of Runge–Kutta-Based Partial Time-Splitting Schemes for the Euler Equations

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ABSTRACT

For atmospheric simulation models with resolutions from about 10 km to the subkilometer cloud-resolving scale, the complete nonhydrostatic compressible Euler equations are often used. An important integration technique for them is the time-splitting (or split explicit) method. This article presents a comprehensive numerical stability analysis of Runge–Kutta (RK)-based partial time-splitting schemes. To this purpose a linearized two-dimensional (2D) compressible Euler system containing advection (as the slow process), sound, and gravity wave terms (as fast processes) is considered. These processes are the most important ones in limiting stability. First, the detailed stability properties are discussed with regard to several off-centering weights for each fast process described by horizontally explicit, vertically implicit schemes. Then the stability properties of the temporally and spatially discretized three-stage RK scheme for the complete 2D Euler equations and their stabilization (e.g., by divergence damping) are discussed. The main goal is to find optimal values for all of the occurring numerical parameters to guarantee stability in operational model applications. Furthermore, formal orders of temporal truncation errors for the time-splitting schemes are calculated. With the same methodology, two alternatives to the three-stage RK method, a so-called RK3-TVD method, and a new four-stage, second-order RK method are inspected.

1. Introduction

Atmospheric motions can occur on a broad range of temporal and spatial scales and therefore are an excellent example of a multiscale system. This is particularly the case if they are described by the complete nonhydrostatic, compressible Euler equations, in which no filtering of fast processes takes place. Today, this equation set is often used for mesoscale model applications with (horizontal) resolutions from 10 km to the subkilometer cloud-resolving scale. Its numerical treatment has to consider the multiscale properties while remaining efficient. One basic strategy to treat the efficiency problem is to use time-split (sometimes called split explicit) schemes. Time-split schemes separate the equations into slow and fast processes and integrate them with large and multiple small time steps, respectively. Efficiency is increased if the slow processes are computationally expensive whereas the fast processes are computationally cheap. On the mesoscale, advection can be considered a slow process, for which higher-order advection schemes are preferred for accuracy reasons. One of the fast processes is the propagation of sound waves. They are meteorologically less important and can therefore be treated with a lower-order scheme.

The so-called complete operator splitting schemes (Durran 1998) have been largely analyzed in the literature [e.g., Skamarock and Klemp (1992); LeVeque and Oliger (1983) denote it as ''additive splitting,'' which should not be mistaken for the ''addition of tendencies'' mentioned later on]. Generally operator splitting schemes are well suited, if the temporal scales of the processes are of comparable magnitude. When equal time steps are used for each subprocess then it is the standard fractional-step splitting. However, Purser and Leslie (1991) have shown that complete operator splitting should not be used to integrate the Euler equations on very different time scales because the coupling between slow and fast processes is too weak. This is particularly the case in more complex flow situations (Durran 1998, see section 7.3.2).

A significant step forward using the time-splitting technique for nonhydrostatic, compressible models has

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been achieved by Klemp and Wilhelmson (1978) and is called partial operator splitting (Durran 1998). Their underlying time-integration method was the three-time-level leapfrog scheme. Partial operator splitting couples the fast and slow processes in a closer manner than complete operator splitting. To extend this method to two time-level schemes, Wicker and Skamarock (1998, 2002) have introduced the Runge–Kutta (RK) method, which allows the use of higher-order upwind advection schemes. In particular, the three-stage RK scheme (RK3WS), which was developed originally for the Advanced Research Weather Research and Forecasting model (WRF; Skamarock and Klemp 2008), is now also implemented into a newer version of the Consortium for Small-Scale Modelling (COSMO) model [formerly known as the Lokal Modell (LM); Doms and Schättler 2002; Steppeler et al. 2003]. Since April 2007 this COSMO version with RK3WS has run at the German Weather Service [Deutscher Wetterdienst (DWD), in a configuration called COSMO-DE], and also at some other European weather services that are members of the COSMO consortium. Those models are running with horizontal grid spacing from 7 km down to the convection-permitting range of about 2 km.

An important property of any discretization scheme is numerical stability. Whereas for nonlinear systems general statements are often impossible, at least linear stability should be investigated when developing and running a model. A very comprehensive stability analysis for the leapfrog-based splitting scheme has been performed by Skamarock and Klemp (1992). They have treated the spatial discretization as exact, as is done in other stability studies (e.g., Gassmann and Herzog 2007; Viner and Epifanio 2008). Such a semidiscrete analysis has the advantage that it can provide analytical stability constraints. In contrast, the von Neumann analysis presented here is performed for the completely (spatially and temporally) discretized equations. This is motivated by Baldauf (2002), who found that for a simple Euler-forward time splitting, a space–time discretization has a stable range at least for small sound Courant numbers, while the semidiscrete analysis shows complete instability. Another well-known example is the simple upwind advection scheme: the unstable Euler-forward time integration is stabilized by an upwind space discretization. It is demonstrated here that at least for subsystems analytical stability constraints can also be derived for the complete space–time analysis.

In this study the two-dimensional (2D), nonhydrostatic, and compressible Euler equations without the Coriolis force are inspected. This system, which describes advection, sound, and gravity wave propagation, contains the most important processes that limit numerical stability of a dynamical core applied to the mesoscale (up to the cloud-resolving or convective scale). In contrast, the stability analysis of Viner and Epifanio (2008) concentrates on large-scale motions and neglects advection. The 2D Euler system also allows us to analyze the vertical discretization, for which atmospheric models usually use relatively small grid spacing on the order of 10 m near the ground. Hence, implicit schemes of a Crank–Nicholson type must be used for the vertical not to restrict the small time step for too much. The resulting tridiagonal systems of equations can be solved nearly as efficient as an explicit scheme. One main goal of this analysis is to find optimal ranges for the off-centering parameters occurring in these tridiagonal systems.

Time-splitting schemes generally need additional damping mechanisms for stabilization. In particular, artificial divergence damping has been proposed as a suitable method by Skamarock and Klemp (1992). Gassmann and Herzog (2007) found that this should be done in an isotropic manner (i.e., that the vertical direction should be damped with the same diffusion coefficient as the horizontal ones). Therefore isotropic divergence damping is used here, too. A further purpose of this study is to find suitable limits for its off-centering parameters and its diffusion coefficient.

Generally, time-splitting schemes are considered here in which each RK substep contains both fast and slow terms and in which the RK scheme has the additional function to stabilize the advection operator. This is close to the original RK3WS. Two other methods are inspected here which share this property: a three-stage RK with a total variation diminishing (TVD) property (Shu 1988; Förstner and Doms 2004) and a new proposal for a time-splitting based on the so-called RK4MS scheme (Baldauf 2008). There also exist slightly different time-splitting approaches that try to avoid artificial divergence damping. Gassmann and Herzog (2007) integrate the fast processes alone in the first substep of a two-stage RK scheme and add a stable advection scheme only in the second RK substep. Recently Jebens et al. (2009) published a time-splitting scheme based on peer methods instead of RK schemes.

This paper is organized as follows. After some general considerations about splitting schemes, section 2 delivers a framework for the analysis of RK-based time-split systems to inspect linear stability properties and the temporal order of truncation error. Section 3 presents the different horizontally explicit, vertically implicit discretizations for sound expansion terms, buoyancy, and artificial divergence damping. Analytical stability constraints are derived from which the parameter space of the several off-centering parameters can be reduced. In section 4 the most important stability constraints and optimal values

for the numerical parameters for meteorological models are inspected by using the compressible 2D Euler equations. Finally, the RK3-TVD method (section 4b) and the new RK4MS scheme (section 4c) as a promising alternative to RK3WS are investigated.

2. Time-splitting schemes

a. General considerations about operator splitting

We consider a general system of PDEs for a vector of fields $\mathbf{q}$, which is composed of two processes:

$$\frac{\partial}{\partial t}\mathbf{q} = \mathcal{P}_s\mathbf{q} + \mathcal{P}_f\mathbf{q}.$$

The indices s and f at the operators (continuous operators are denoted by calligraphic letters) denote "slow" and "fast" processes, respectively. Examples for fast processes $\mathcal{P}_f$ are sound or gravity wave expansion and advection as a slow process $\mathcal{P}_s$. If the two (spatial) operators $\mathcal{P}_s$ and $\mathcal{P}_f$ are not explicitly dependent on time t then the formal solution is $\mathbf{q}(t+\Delta T) = e^{\Delta T(\mathcal{P}_s+\mathcal{P}_f)}\mathbf{q}(t)$. For generally noncommuting operators $\mathcal{P}_s$ and $\mathcal{P}_f$ the exponential can be expanded into

$$e^{\Delta T(\mathcal{P}_s+\mathcal{P}_f)} = 1 + \Delta T(\mathcal{P}_s + \mathcal{P}_f) + \frac{\Delta T^2}{2!}(\mathcal{P}_s^2 + \mathcal{P}_s\mathcal{P}_f + \mathcal{P}_f\mathcal{P}_s + \mathcal{P}_f^2) + O(\Delta T^3). \quad (1)$$

Now we consider discretizations for the operators $\mathcal{P}$ in the framework of *two time-level schemes*, which integrate the state variables $\mathbf{q}$ from time step t^n to $t^{n+1} = t^n + \Delta T$ (time indices are denoted as upper indices). Then the whole scheme is written formally as

$$\mathbf{q}^{t+\Delta T} = Q\Delta T\mathbf{q}^t$$

[discretized (also nonlinear) operators (e.g., Q and identity operator 1) are denoted in italics]. It will be the task of the remaining sections to calculate the eigenvalues of Q to determine the stability properties. To prepare this, in this section Q will be expressed by the slow and fast process operators for RK-based time-splitting schemes. Additionally its temporal truncation error will be considered here. To start with, discretized operators are introduced in a similar formal way for the slow process $\mathcal{P}_s$ alone:

$$\mathbf{q}^{t+\Delta T} = Q_{s,\Delta T}\mathbf{q}^t \equiv Q_s\mathbf{q}^t$$

and for $\mathcal{P}_f$ alone:

$$\mathbf{q}^{t+\Delta t} = Q_{f,\Delta t}\mathbf{q}^t \equiv Q_f\mathbf{q}^t.$$

In the following the numerical operator Q_s advances the fields with a large time step ΔT whereas Q_f always advances with a small time step Δt. In general stability constraints for the fast process will allow only smaller time steps Δt compared to ΔT, if it is discretized by an explicit and thus not unconditionally stable scheme. For this purpose, a *time-splitting ratio* $n_s := \Delta T/\Delta t$ is defined and chosen as an integer number.

We consider time integration schemes that deliver one large time step ΔT as composed of the single processes. Of course, these schemes have to be consistent (i.e., at least of first order in time). To derive the temporal order we consider the following expansions:

$$Q_f = 1 + \Delta t P_f^{(1)} + \frac{\Delta t^2}{2!}P_f^{(2)} + \frac{\Delta t^3}{3!}P_f^{(3)} + O(\Delta t^4) \quad (2)$$

and

$$Q_s = 1 + \Delta T P_s^{(1)} + \frac{\Delta T^2}{2!}P_s^{(2)} + \frac{\Delta T^3}{3!}P_s^{(3)} + O(\Delta T^4). \quad (3)$$

For example, in an Euler-forward scheme for the slow processes, only $P_s^{(1)}$ is used, whereas all the other $P_s^{(2)} = P_s^{(3)} = \cdots = 0$; in a Lax–Wendroff discretization $P_s^{(1)}$ and $P_s^{(2)}$ are used, whereas $P_s^{(3)} = P_s^{(4)} = \cdots = 0$. Despite the literal similarity between the analytic operator $\mathcal{P}$ and the numerical operators $P_s^{(1)}$, $P_s^{(2)}, \ldots$ one should observe that even for a second-order scheme in general

$$P_s^{(2)} \neq (P_s^{(1)})^2, \quad (4)$$

in contrast to the analytic expansion in Eq. (1). The reason for this ambiguity is the freedom in the spatial discretization, which can be of arbitrary order. It is only required that for a consistent scheme it must hold that

$$P_s^{(1)} \to \mathcal{P}_s, \quad (5)$$

if the spatial grid spacing converges to zero (e.g., $\Delta x \to 0$, $\Delta y \to 0, \ldots$). For a second-order scheme (in time) it must hold additionally that

$$P_s^{(2)} \to \mathcal{P}_s^2, \quad (6)$$

and for a third-order scheme additionally that $P_s^{(3)} \to \mathcal{P}_s^3$, and so on. The same is valid, of course, for the fast processes.

Similarly, the conditions for the whole scheme can be derived. The minimal condition for any discretization is consistency, which is fulfilled if

$$Q = 1 + \Delta T P_s^{(1)} + n_s\Delta t P_f^{(1)} + O(\Delta T^2). \quad (7)$$

One cannot write down in the same explicit manner the conditions for second-order accuracy, due to the ambiguities as in Eq. (4). But if one carries out the spatial limit $\Delta x \to 0$, $\Delta y \to 0, \ldots$, and obeys Eqs. (5) and (6) and the related limits for $P_f^{(1)}$ and $P_f^{(2)}$, respectively, then the second-order expression of Eq. (1) must follow.

b. Time-split Euler-forward scheme

The simplest scheme that combines slow and fast processes in a closer manner than complete operator splitting is the partially split Euler-forward (EF) scheme. In a first step the tendency of the slow processes is calculated by integrating $\mathbf{q}$ over ΔT:

$$\mathbf{q}^* = Q_s \mathbf{q}^n.$$

This generates the slow tendency:

$$\left.\frac{d\mathbf{q}}{dt}\right|_s := \frac{\mathbf{q}^* - \mathbf{q}^n}{\Delta T} = \frac{1}{\Delta T}(Q_s - 1)\mathbf{q}^n. \tag{8}$$

In the next step one again starts from the state $\mathbf{q}^n$ but now the fast processes are integrated with a small time step Δt (the appropriate time level is marked with a second upper index), and in every step the slow tendency is added:

$$\mathbf{q}^{n,1} = Q_f \mathbf{q}^n + \Delta t \left.\frac{d\mathbf{q}}{dt}\right|_s = Q_f \mathbf{q}^n + \frac{1}{n_s}(Q_s - 1)\mathbf{q}^n, \tag{9}$$

$$\mathbf{q}^{n,2} = Q_f \mathbf{q}^{n,1} + \Delta t \left.\frac{d\mathbf{q}}{dt}\right|_s = Q_f^2 \mathbf{q}^n + (Q_f + 1)\frac{Q_s - 1}{n_s}\mathbf{q}^n, \tag{10}$$

and so on until $\mathbf{q}^{n,ns} \equiv \mathbf{q}^{n+1}$ is achieved. Summarized this results in

$$\mathbf{q}^{n+1} = Q_{\mathrm{EF}} \mathbf{q}^n$$

with the time-split Euler-forward operator:

$$Q_{\mathrm{EF}} := Q_f^{n_s} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s-1}(Q_f)^i\right](Q_s - 1). \tag{11}$$

The practical computation amount per large time step is approximately $1\tau_s + n_s\tau_f$, in which τ_s and τ_f are the wall clock times to calculate the slow and fast processes, respectively. Skamarock and Klemp (1992) carried out a detailed stability analysis for this scheme by treating the spatial operators as exact and found that this scheme is unconditionally unstable. If the analysis is extended also to include the spatial discretizations, then stabilization at least for smaller sound Courant numbers can be achieved by the use of a strong divergence damping (Baldauf 2002). Nevertheless, the stability region is small and in general this scheme is not well suited to handle higher-order advection operators in a stable manner. However, in meteorological models (also in the COSMO model) partially split EF is sometimes used to couple physics parameterizations and the dynamical core (e.g., to use a longer time step for convection parameterizations). The temporal order of this scheme is just Eq. (7), but never exceeds one.

c. Time-split three-stage Runge–Kutta scheme

The use of a three-stage RK scheme [to solve an ODE $dq/dt = f(q)$] of the following form:

$$\begin{aligned} q^* &= q^n + \Delta t \frac{1}{3} f(q^n), \\ q^{**} &= q^n + \Delta t \frac{1}{2} f(q^*), \\ q^{n+1} &= q^n + \Delta t f(q^{**}) \end{aligned} \tag{12}$$

in time-splitting schemes has been proposed by Wicker and Skamarock (2002). As mentioned in Baldauf (2008), it is only second-order accurate for nonlinear equations but becomes third-order accurate for linear problems. Hence, it has been denoted there as a "linear case RK scheme," for which a general stability theory has been derived.

The derivation of an operator for this RK3WS time-splitting scheme is similar to the EF scheme: the first RK step is a time-split EF scheme starting from $\mathbf{q}^n$ over the time interval $\Delta T/3$ using $n_s/3$ small time steps ending up with $\mathbf{q}^* = Q_{\mathrm{RK3WS,1}}\mathbf{q}^n$ with the following operator:

$$Q_{\mathrm{RK3WS,1}} := Q_f^{n_s/3} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/3-1}(Q_f)^i\right](Q_s - 1).$$

The second RK step is a time-split EF scheme starting from $\mathbf{q}^n$ over the time interval $\Delta T/2$ using $n_s/2$ small time steps. But in every small time step there is added a corrected slow tendency

$$\left.\frac{d\mathbf{q}}{dt}\right|_{s,2} := \frac{Q_s\mathbf{q}^* - \mathbf{q}^*}{\Delta T} = \frac{Q_s - 1}{\Delta T} Q_{\mathrm{RK3WS,1}}\mathbf{q}^n,$$

which is determined by starting from $\mathbf{q}^*$. This results in

$$Q_{\mathrm{RK3WS,2}} := Q_f^{n_s/2} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/2-1}(Q_f)^i\right](Q_s - 1)Q_{\mathrm{RK3WS,1}}.$$

The third RK step can be derived in the same manner and leads to the complete operator:

$$Q_{\mathrm{RK3WS}} := Q_f^{n_s} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s-1}(Q_f)^i\right](Q_s - 1)Q_{\mathrm{RK3WS},2}. \quad (13)$$

The computation amount per large time step is approximately $3\tau_s + (11/6)n_s\tau_f$.

To perform a truncation error analysis the expansions Eqs. (2) and (3) are inserted into Q_{RK3WS} and expanded by powers of ΔT. After some tedious calculations this yields

$$\begin{aligned} Q_{\mathrm{RK3WS}} = 1 + \Delta T(P_s^{(1)} + P_f^{(1)}) + \frac{\Delta T^2}{2}\Bigg[& P_s^{(2)} + (P_s^{(1)})^2 \\ & + P_s^{(1)}P_f^{(1)} + \left(1 - \frac{1}{n_s}\right)P_f^{(1)}P_s^{(1)} + \left(1 - \frac{1}{n_s}\right)(P_f^{(1)})^2 \\ & + \frac{1}{n_s}P_f^{(2)}\Bigg] + O(\Delta T)^3. \end{aligned} \quad (14)$$

The comparison with the analytic operator in Eq. (1) shows that the scheme is consistent (first order in time) if the single processes are at least of first order, that is, $P_f^{(1)} \to \mathcal{P}_f$, $P_s^{(1)} \to \mathcal{P}_s$ for all spatial grid spacing tending to zero ($\Delta x \to 0, \ldots$), whereas the higher-order operators $P_f^{(2)}$ and $P_s^{(2)}$ do not have to fulfill any constraints. They could be zero as for EF schemes. One further sees that if both processes are discretized in higher order in time then the term $\mathcal{P}_s^2$ arises with a false factor of 2 instead of 1. The reason is that RK schemes are designed to combine pure EF steps to a correct scheme of higher order. Therefore, a minimal requirement toward higher order is to calculate the slow process as a true Euler-forward scheme. Or to be more precise: with a discretization that has at least $P_s^{(2)} \equiv 0$. This is the case for the advection operators used in Wicker and Skamarock (1998, 2002). It is not valid, for example, for slow processes that are partially calculated by higher-order schemes, such as Lax–Wendroff advection or vertically implicit schemes. Indeed, the WRF model (Klemp et al. 2007) uses an EF advection scheme also in the vertical. However, a stability restriction by smaller vertical grid spacing can occur. In the COSMO model vertically implicit advection is used to circumvent this stability constraint. But obviously this reduces the temporal order even of higher-order RK time-integration schemes. From this a general recommendation can be derived to use implicit schemes for slow process integration only outside of the RK scheme.

If the fast processes are at least of second order in Δt (e.g., forward–backward or Crank–Nicholson; see below) then this RK3WS scheme is of second order in ΔT *in the limiting case* $n_s \to \infty$. Inspecting the third-order terms (not shown) one finds that even for $n_s \to \infty$, $P_s^{(2)} = 0$, $P_s^{(3)} = 0$, and even for commuting operators $P_s^{(1)}$ and $P_f^{(1)}$, it is never a third-order scheme in ΔT.

3. Discretizations for the linearized 2D Euler equations

In the following we want to inspect stability properties of the different time-splitting schemes. To this purpose the 2D, linearized, nonhydrostatic, and compressible Euler equations

$$\frac{\partial u}{\partial t} + U_0\frac{\partial u}{\partial x} = -\frac{1}{\rho_0}\frac{\partial p'}{\partial x} + \alpha_{\mathrm{div}}\frac{\partial D}{\partial z}, \quad (15)$$

$$\frac{\partial w}{\partial t} + U_0\frac{\partial w}{\partial x} = -\frac{1}{\rho_0}\frac{\partial p'}{\partial z} + g\left(\frac{T'}{T_0} - \frac{p'}{p_0}\right) + \alpha_{\mathrm{div}}\frac{\partial D}{\partial z}, \quad (16)$$

$$\frac{\partial p'}{\partial t} + U_0\frac{\partial p'}{\partial x} = -\frac{c_p}{c_v}p_0 D + \rho_0 g w, \quad \text{and} \quad (17)$$

$$\underbrace{\frac{\partial T'}{\partial t} + U_0\frac{\partial T'}{\partial x}}_{-\mathcal{P}_A} = \underbrace{-\frac{R}{c_v}T_0 D}_{\mathcal{P}_S} \underbrace{-\frac{\partial T_0}{\partial z}w}_{\mathcal{P}_B} \underbrace{}_{\mathcal{P}_D}, \quad (18)$$

with the divergence

$$D = \frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} \quad (19)$$

are considered. Here terms are grouped as indicated into the processes "advection" $\mathcal{P}_A$, "sound propagation" $\mathcal{P}_S$, "buoyancy" $\mathcal{P}_B$, and a possible artificial divergence damping term $\mathcal{P}_D$. It should be emphasized that the partitioning into sound and buoyancy is motivated only numerically to separately determine stability limits for several occurring numerical weighting parameters. Whereas $\mathcal{P}_S$ correctly describes the expansion of sound waves, physically significant gravity waves only occur as a result of the interaction of both sound and buoyancy processes.

This vertical slice model contains the most important features that restrict the stability of spatial and temporal discretization schemes of a nonhydrostatic 3D model applied to the convection permitting scale. We write this Euler equation system formally as

$$\frac{\partial \mathbf{q}}{\partial t} = \mathcal{P}_A\mathbf{q} + \mathcal{P}_S\mathbf{q} + \mathcal{P}_B\mathbf{q} + \mathcal{P}_D\mathbf{q} \quad (20)$$

with the vector for the primitive variables $\mathbf{q} := (u, w, p', T')$, where $p = p_0 + p'$ and $T = T_0 + T'$.

For the derivation of the buoyancy term in the equation for vertical velocity a hydrostatic and steady base state (p_0, T_0, ρ_0, U_0) has been introduced (Steppeler et al. 2003; Doms and Schättler 2002) and for small perturbations it can be linearized to

$$-g\frac{\rho'}{\rho} = g\frac{\rho_0}{\rho}\left(\frac{T'}{T} - \frac{T_0}{T}\frac{p'}{p_0}\right) \approx g\left(\frac{T'}{T_0} - \frac{p'}{p_0}\right).$$

As long as the base-state fields arise as coefficients they are assumed constant, but constant gradients $\partial T_0/\partial z$ and $\partial p_0/\partial z = -g\rho_0$ (again as coefficient functions) are allowed. This little and commonly used inconsistency is allowed insofar as the changes in the fields over a vertical distance L_z are small, that is, $|\partial T_0/\partial z|L_z \ll T_0$ and $L_z \ll 1/\delta$ with the inverse density-scale height $\delta := -1/\rho_0 \partial\rho_0/\partial z$. This obviously limits the range of applicability of the analysis to vertical motions with extensions L_z much less than about 10 km. This restriction could be weakened by analyzing systems that use different prognostic variables such as potential temperature or that perform a Bretherton transformation (BT) to reduce the vertical variability of the coefficient functions (Gassmann and Herzog 2007). Indeed, Eqs. (15)–(19) can be considered as a BT system (with constant coefficients $\rho_0, \ldots$), in which correction terms proportional to $\delta/2$ are neglected; this leads to a slightly weaker condition $L_z \ll 4\pi/\delta$. In appendix B it is shown that the combination of a BT (*with* $\delta \neq 0$) and the artificial divergence damping leads to slightly unstable waves even in the pure analytic case. This is one reason why the equations in the above form are used as a starting point for the numerical stability analysis. Furthermore, no boundaries are considered; instead, waves can expand infinitely. This additionally legitimizes the above mentioned restriction of the vertical scales: only waves should be considered that are far away from the top or bottom boundaries (i.e., that are small compared to the vertical extension of the model). Other restrictions of this analysis are the neglect of any metric terms that arise by using a terrain-following coordinate, and the absence of vertical advection.

Beyond this, there is currently a general trend in atmospheric modeling to use density ρ as a prognostic variable, rather than pressure p, to better control mass conservation. In contrast, here we want to be as close as possible to the $p'T'$ equation system used in the COSMO model.

To carry out a von Neumann stability analysis (for details see, e.g., Baldauf 2008) we introduce an expansion into waves [lower indices denote the spatial discretization on a 2D (horizontal and vertical) equidistant grid]:

$$u^n_{j,k} = u^n(k_x, k_z)e^{i(k_x j\Delta x + k_z k\Delta z)},$$

analogous for $w^n_{j,k}$, $T'^n_{j,k}$ and $p'^n_{j,k}$. Then the numerical operators Q are matrices with a block diagonal form, in which each 4×4 block $\mathbf{Q}$ refers to a particular wave vector (k_x, k_z). Finally the eigenvalues of the resulting matrix equations are determined. Stability holds if the absolute values of all eigenvalues are bounded by 1. Therefore, only the spectral radius of the amplification matrix is determined and not the spectral norm. However, the spectral radius is in general a good indicator for stability as discussed in the appendix of Skamarock and Klemp (1992). For convenience a vector of fields $\tilde{\mathbf{q}}^n(k_x, k_z) := (\tilde{u}^n, \tilde{w}^n, \tilde{p}'^n, \tilde{T}'^n)^{\mathrm{T}}$ is introduced. Here the fields are made dimensionless by the sound velocity $c_s := \sqrt{c_p/c_v}\sqrt{p_0/\rho_0}$ (with the specific heats of dry air at constant pressure c_p or volume c_v) or by the base-state values p_0 and T_0, respectively: $\tilde{u}^n := u^n(k_x, k_z)/c_s$, $\tilde{w}^n := w^n(k_x, k_z)/c_s$, $\tilde{p}'^n := p'^n(k_x, k_z)/p_0$, and $\tilde{T}'^n := T'^n(k_x, k_z)/T_0$.

Of course, a numerical stability analysis makes sense only for those physical states that are not themselves unstable. The base state considered here, a nonshear flow in a stably stratified atmosphere, fulfills this condition.

Before considering discretizations of the complete system, in the next sections we discuss distinct discretizations of the slow and fast processes separately.

a. Advection term discretization

A discussion of stability properties of several advection terms with a comprehensive theory of RK schemes is presented in Baldauf (2008). Here it means to define a 4×4 matrix $\mathbf{Q}^{\mathrm{adv}}$ describing the horizontal advection,

$$\tilde{\mathbf{q}}^{n+1} = \mathbf{Q}^{\mathrm{adv}}\tilde{\mathbf{q}}^n,$$

in which

$$\mathbf{Q}^{\mathrm{adv}} = -iC_{\mathrm{adv}}d(K)\cdot\mathbf{1} \tag{21}$$

is a multiple of the identity and $C_{\mathrm{adv}} = U_0\Delta T/\Delta x$ is the advection Courant number. The effective (dimensionless) wavenumber $d(K)$ is described in Baldauf (2008) [see his Eqs. (35)–(40)] for upwind or centered difference discretizations of first to sixth order. In particular the fifth-order upwind scheme has

$$d(K) = \frac{i}{60}(2e^{-3iK} - 15e^{-2iK} + 60e^{-iK} - 20 - 30e^{iK} + 3e^{2iK}), \tag{22}$$

with the horizontal dimensionless wavenumber $K := k_x\Delta x$. In combination with "linear case" RK3 methods this scheme is stable for advection Courant numbers $C_{\mathrm{adv}} \leq 1.435$.

b. Sound terms

In this section we only inspect terms responsible for (linear) sound expansion:

$$\frac{\partial \mathbf{q}}{\partial t} = \mathcal{P}_S \mathbf{q}.$$

To this purpose a completely generalized Crank–Nicholson scheme with a variety of different off-centering coefficients β_i^s for each term is considered:

$$\begin{aligned}
\frac{u^{n+1}-u^n}{\Delta t} &= -\frac{1}{\rho_0}[\beta_1^s \delta_x p'^{n+1} + (1-\beta_1^s)\delta_x p'^n],\\
\frac{w^{n+1}-w^n}{\Delta t} &= -\frac{1}{\rho_0}[\beta_2^s \delta_z p'^{n+1} + (1-\beta_2^s)\delta_z p'^n],\\
\frac{p'^{n+1}-p'^n}{\Delta t} &= -\frac{c_p}{c_v} p_0[\beta_3^s \delta_x u^{n+1} + (1-\beta_3^s)\delta_x u^n\\
&\quad + \beta_4^s \delta_z w^{n+1} + (1-\beta_4^s)\delta_z w^n],\\
\frac{T'^{n+1}-T'^n}{\Delta t} &= -\frac{R}{c_v} T_0[\beta_5^s \delta_x u^{n+1} + (1-\beta_5^s)\delta_x u^n\\
&\quad + \beta_6^s \delta_z w^{n+1} + (1-\beta_6^s)\delta_z w^n].
\end{aligned}$$

Here $\beta_i^s = 0$ is a fully explicit term and $\beta_i^s = 1$ is a fully implicitly treated term. The "pure" second-order Crank–Nicholson (trapezoidal) scheme results for $\beta_i^s = ½$. Here a staggered Arakawa–C–Lorenz-grid is used, in which scalars p and T are defined at the position j, k, and the vector components at staggered positions: $u_{j+1/2,k}$, $w_{j,k+1/2}$. For all the spatial discretizations centered differences

$$(\delta_x \phi)_{jk} = \frac{\phi_{j+1/2,k} - \phi_{j-1/2,k}}{\Delta x}$$

are used, analogously for δ_z.

The stability analysis leads to

$$\begin{aligned}
(\mathbf{1} + \mathbf{P}_1^{\mathrm{snd}})\tilde{\mathbf{q}}^{n+1}(k_x, k_z) &= (\mathbf{1} + \mathbf{P}_2^{\mathrm{snd}})\tilde{\mathbf{q}}^{n}(k_x, k_z) \quad \text{or}\\
\tilde{\mathbf{q}}^{n+1} &= \mathbf{Q}^{\mathrm{snd}}\tilde{\mathbf{q}}^{n} \quad \text{with}\\
\mathbf{Q}^{\mathrm{snd}} &:= (\mathbf{1} + \mathbf{P}_1^{\mathrm{snd}})^{-1} \cdot (\mathbf{1} + \mathbf{P}_2^{\mathrm{snd}}),
\end{aligned} \tag{23}$$

with the appropriate matrices

$$\mathbf{P}_1^{\mathrm{snd}} = 2i \begin{pmatrix} 0 & 0 & \frac{c_v}{c_p} S_x \beta_1^s & 0\\ 0 & 0 & \frac{c_v}{c_p} S_z \beta_2^s & 0\\ \frac{c_p}{c_v} S_x \beta_3^s & \frac{c_p}{c_v} S_z \beta_4^s & 0 & 0\\ \frac{R}{c_v} S_x \beta_5^s & \frac{R}{c_v} S_z \beta_6^s & 0 & 0 \end{pmatrix} \tag{24}$$

and

$$\mathbf{P}_2^{\mathrm{snd}} = -2i \begin{bmatrix} 0 & 0 & \frac{c_v}{c_p} S_x (1-\beta_1^s) & 0\\ 0 & 0 & \frac{c_v}{c_p} S_z (1-\beta_2^s) & 0\\ \frac{c_p}{c_v} S_x (1-\beta_3^s) & \frac{c_p}{c_v} S_z (1-\beta_4^s) & 0 & 0\\ \frac{R}{c_v} S_x (1-\beta_5^s) & \frac{R}{c_v} S_z (1-\beta_6^s) & 0 & 0 \end{bmatrix}. \tag{25}$$

Here the abbreviations

$$S_x := C_{\mathrm{snd},x} \sin\frac{k_x \Delta x}{2}, \quad S_z := C_{\mathrm{snd},z} \sin\frac{k_z \Delta z}{2}$$

are used. Furthermore, *horizontal* and *vertical sound-Courant numbers* $C_{\mathrm{snd},x} := c_s \Delta t/\Delta x$ and $C_{\mathrm{snd},z} := c_s \Delta t/\Delta z$ are defined, respectively, both with the *small* time step Δt. In COSMO-DE $\Delta t \approx 5$ s is used, which leads to $C_{\mathrm{snd},x} \approx 340\ \mathrm{m\ s^{-1}} \cdot 5\ \mathrm{s}/(2800\ \mathrm{m}) \approx 0.6$.

The determination of the eigenvalues of $\mathbf{Q}^{\mathrm{snd}}$ leads to the following (partially familiar) stability statements (see the first section of appendix A for a more detailed analysis). At first, the characteristic polynomial $\det(\mathbf{Q}^{\mathrm{snd}} - \lambda \cdot \mathbf{1})$ is independent of β_5^s and β_6^s. The reason is that sound propagation only stems from the coupling of the pressure gradient terms in the momentum equations and the divergence term in the pressure equation. Hence the stability of any numerical discretization is determined only by these three equations and not by the temperature equation. Therefore, looking to the stability of sound expansion alone, β_5^s and β_6^s could be chosen arbitrarily. But probably these two parameters will get importance when coupling sound with other processes. In the following they are set analogously to the pressure equation: $\beta_5^s = \beta_3^s$ and $\beta_6^s = \beta_4^s$.

The case of a fully explicit discretization is unconditionally unstable. The reason is the use of spatial centered difference discretizations in an Euler-forward time stepping. This fact led Mesinger (1977) to the use of *forward–backward integration*, which can be done in the two variants: $\{\beta_1^s = \beta_2^s = 0, \beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = 1\}$ or $\{\beta_1^s = \beta_2^s = 1, \beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = 0\}$. Both variants are *stable and neutral* for $C_x^2 + C_z^2 \le 1$. As stated in Mesinger (1977), though the forward equation is only of first order, the scheme as a whole is of second order in time.

Because of small vertical grid spacing near the bottom (usually $\Delta z = 20$ m for the COSMO models), $C_{\mathrm{snd},z}$ can easily obtain values much bigger than 1. Therefore the forward–backward scheme in the horizontal is combined with a vertically implicit scheme. This leads to $\beta_1^s = 0$

(forward calculation of the u equation) and $\beta_3^s = 1, \beta_5^s = 1$ (backward calculation of the p and T equations). This is sufficient to lead to a tridiagonal system of equations and therefore is efficient; β_2^s, β_4^s, and β_6^s determine the degree of implicitness. In the special case $\beta_2^s = \beta_4^s =: \beta^s$ the scheme is stable for $|C_x| \leq 1, ½ \leq \beta^s \leq 1$, and arbitrary C_z. For $\beta^s > ½$ it is damping, whereas for a pure Crank–Nicholson variant, that is, $\beta_2^s = \beta_4^s = ½$, it is neutral for all S_z and $|S_x| \leq 1$.

c. *Buoyancy terms*

The buoyancy terms [see Eq. (20)]

$$\frac{\partial \mathbf{q}}{\partial t} = \mathcal{P}_B \mathbf{q}$$

directly lead to an oscillation equation

$$\frac{\partial^2 w}{\partial t^2} + \omega_a^2 w = 0$$

with the Brunt–Väisälä-frequency

$$N^2 = \frac{g}{T_0}\left(\frac{g}{c_p} + \frac{\partial T_0}{\partial z}\right)$$

and the acoustic cutoff frequency

$$\omega_a^2 = N^2 + \frac{g^2}{c_s^2}.$$

For commonly occurring temperature stratification the value of $\omega_a \approx 0.03\ \mathrm{s}^{-1}$ is dominated by the second term on the rhs.

Again we discretize by a generalized Crank–Nicholson scheme. In a staggered grid the averaging operator

$$(A_z\phi)_{jk} = \frac{1}{2}(\phi_{j,k+1/2} + \phi_{j,k-1/2})$$

is used (analogous for the fields u and w on staggered positions). This leads to

$$\frac{w^{n+1} - w^n}{\Delta t} = g\left[\beta_1^B A_z \frac{T'^{n+1}}{T_0} + (1-\beta_1^B) A_z \frac{T'^n}{T_0} - \beta_2^B A_z \frac{p'^{n+1}}{p_0} - (1-\beta_2^B) A_z \frac{p'^n}{p_0}\right],$$

$$\frac{p'^{n+1} - p'^n}{\Delta t} = \rho_0 g[\beta_3^B A_z w^{n+1} + (1-\beta_3^B) A_z w^n], \quad \text{and}$$

$$\frac{T'^{n+1} - T'^n}{\Delta t} = -\frac{\partial T_0}{\partial z}[\beta_4^B A_z w^{n+1} + (1-\beta_4^B) A_z w^n].$$

We remark that for any choice of the off-centering weights β_1^B, β_2^B, β_3^B, and β_4^B these equations alone lead at most to a tridiagonal system for $w_{j,k+3/2}^{n+1}$, $w_{j,k+1/2}^{n+1}$, and $w_{j,k-1/2}^{n+1}$.

One can introduce two independent dimensionless parameters, at which

$$C_g := \frac{g\Delta t}{c_s}$$

is a dimensionless time step, whereas

$$C_\beta := \frac{1}{T_0}\frac{\partial T_0}{\partial z}\frac{c_s^2}{g}$$

is a measure for the strength of the temperature stratification. For the "standard atmosphere" holds $C_\beta \approx -0.24$, whereas neutral (or dry adiabatic) stratification has $C_\beta = -R/c_v = -287/717 \approx -0.40$. The acoustic cutoff frequency can be expressed by

$$\omega_a \Delta t = C_g \sqrt{\frac{c_p}{c_v} + C_\beta} =: C_{\mathrm{buoy}} \tag{26}$$

and defines a buoyancy Courant number C_{buoy}.

The von Neumann stability analysis leads to

$$(\mathbf{1} + \mathbf{P}_1^{\mathrm{buoy}})\tilde{\mathbf{q}}^{n+1}(k_x, k_z) = (\mathbf{1} + \mathbf{P}_2^{\mathrm{buoy}})\tilde{\mathbf{q}}^{n}(k_x, k_z)$$

with

$$\mathbf{P}_1^{\mathrm{buoy}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & S_g\beta_2^B & -S_g\beta_1^B \\ 0 & -\frac{c_p}{c_v}S_g\beta_3^B & 0 & 0 \\ 0 & C_\beta S_g \beta_4^B & 0 & 0 \end{bmatrix} \tag{27}$$

and

$$\mathbf{P}_2^{\mathrm{buoy}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -S_g(1-\beta_2^B) & S_g(1-\beta_1^B) \\ 0 & +\frac{c_p}{c_v}S_g(1-\beta_3^B) & 0 & 0 \\ 0 & -C_\beta S_g(1-\beta_4^B) & 0 & 0 \end{bmatrix} \tag{28}$$

(here the trivial equation $\partial u/\partial t = 0$ is added) and the abbreviation is

$$S_g := C_g \cos\frac{k_z \Delta z}{2}.$$

Alternatively in a nonstaggered grid the averaging operator A_z is an identity and leads to $S_g = C_g$. Because of $|k_z\Delta z| \leq \pi$ it follows that $S_g \in [0, C_g]$. Therefore every stability constraint for S_g can be written as an equal constraint for C_g. This leads to the fact that the following stability conditions are the same for staggered C grids and unstaggered A grids.

The following paragraphs summarize the stability properties of this discretization; a more detailed analysis is carried out in the second section of appendix A. Again a *fully explicit* scheme is unconditionally unstable.

A *forward–backward* scheme with either $\{\beta_1^B = \beta_2^B = 0 \text{ and } \beta_3^B = \beta_4^B = 1\}$ or $\{\beta_1^B = \beta_2^B = 1 \text{ and } \beta_3^B = \beta_4^B = 0\}$ is stable and neutral, if $C_{\text{buoy}} < 2$, or $\Delta t^2(N^2 + g^2/c_s^2) < 4$, which limits the (small) time step to about 60 s (relatively independent from stratification). This constraint can be compared with $\Delta t^2 N^2 < 1$ found in Skamarock and Klemp (1992) for an equation system using the potential temperature, which restricts the time step to about 100 s for a standard atmosphere and to about 50 s for an isothermal stratification.

The stability analysis for arbitrary β_i^B is quite complicated but in the case of a *generalized Crank–Nicholson scheme* in which all β_i^B are equal $\beta_1^B = \beta_2^B = \beta_3^B = \beta_4^B \equiv \beta^B$, the problem is solvable: for $\beta^B < ½$ the scheme is unconditionally unstable, $\beta^B = ½$ leads to an unconditionally stable and neutral scheme, and for $\beta^B > ½$ the scheme is also unconditionally stable, but damping.

Therefore with the Crank–Nicholson discretization no stability limitation occurs for the pure buoyancy terms and they could in principle be treated as slow processes. But gravity waves in general are quite tightly coupled to sound waves and should be treated within the small time steps as argued in Klemp et al. (2007). Also Viner and Epifanio (2008) find smaller phase errors for a large-scale Rossby wave problem if they treat buoyancy as a fast process. Moreover, the computational costs for buoyancy terms are small, if the implicit part is solved together with the other fast processes.

d. Divergence damping

To stabilize the whole time-integration scheme, a divergence damping has been introduced in Skamarock and Klemp (1992) by adding the gradient of a divergence on the rhs of the momentum equations:

$$\frac{\partial}{\partial t}\mathbf{v} = \alpha_{\text{div}}\nabla(\nabla\cdot\mathbf{v}).$$

They have shown that this additional term generates diffusion for the divergence and therefore damps sound waves, but only slightly reduces the amplitude of the gravity waves. It also drives the compressible system slightly toward an anelastic one.

The discretization on a staggered grid leads to

$$\begin{aligned}\frac{u^{n+1}-u^n}{\Delta t} &= \alpha_{\text{div}}[\beta_1^d\delta_x\delta_x u^{n+1} + (1-\beta_1^d)\delta_x\delta_x u^n \\ &\quad + \beta_2^d\delta_x\delta_z w^{n+1} + (1-\beta_2^d)\delta_x\delta_z w^n], \\ \frac{w^{n+1}-w^n}{\Delta t} &= \alpha_{\text{div}}[\beta_3^d\delta_z\delta_x u^{n+1} + (1-\beta_3^d)\delta_z\delta_x u^n \\ &\quad + \beta_4^d\delta_z\delta_z w^{n+1} + (1-\beta_4^d)\delta_z\delta_z w^n].\end{aligned}$$

A stability analysis now delivers

$$\left(\mathbf{1}+\mathbf{P}_1^{\text{div}}\right)\begin{pmatrix}\tilde{u}^{n+1}\\ \tilde{w}^{n+1}\end{pmatrix} = \left(\mathbf{1}+\mathbf{P}_2^{\text{div}}\right)\begin{pmatrix}\tilde{u}^{n}\\ \tilde{w}^{n}\end{pmatrix},$$

with the matrices

$$\begin{aligned}\mathbf{P}_1^{\text{div}} &= \begin{bmatrix}\beta_1^d S_x S_x & \beta_2^d S_x S_z\\ \beta_3^d S_x S_z & \beta_4^d S_z S_z\end{bmatrix}, \\ \mathbf{P}_2^{\text{div}} &= \begin{bmatrix}-(1-\beta_1^d)S_x S_x & -(1-\beta_2^d)S_x S_z\\ -(1-\beta_3^d)S_x S_z & -(1-\beta_4^d)S_z S_z\end{bmatrix}.\end{aligned} \tag{29}$$

Here divergence Courant numbers

$$C_{\text{div},x} := \alpha_{\text{div}}\frac{\Delta t}{\Delta x^2}, \quad C_{\text{div},z} := \alpha_{\text{div}}\frac{\Delta t}{\Delta z^2}$$

and derivation operators

$$S_x := 2\sqrt{C_{\text{div},x}}\sin\frac{k_x\Delta x}{2}, \quad S_z := 2\sqrt{C_{\text{div},z}}\sin\frac{k_z\Delta z}{2}$$

have been introduced.

Again we determine the eigenvalues of $\mathbf{Q}^{\text{div}} = (\mathbf{1} + \mathbf{P}_1^{\text{div}})^{-1}(\mathbf{1} + \mathbf{P}_2^{\text{div}})$. A fully explicit scheme (all $\beta_i^d = 0$) leads to the eigenvalues $\lambda_1 = 1, \lambda_2 = 1 - (S_x^2 + S_z^2)$ and therefore is stable, if $C_{\text{div},x} + C_{\text{div},z} \leq ½$.

Because of the quadratic factor in $C_{\text{div},z} = (\Delta x/\Delta z)^2 C_{\text{div},x}$, in particular $C_{\text{div},z}$ can achieve large values, at least near the bottom boundary. Hence in any case an unconditionally stable discretization in the vertical direction is needed. A fully implicit scheme (all $\beta_i^d = 1$) has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 1/(1 + S_x^2 + S_z^2)$ and therefore is unconditionally stable. But it does not lead to a tridiagonal system and is therefore inappropriate for split-explicit schemes.

Hence, we apply a horizontally forward–backward, vertically generalized Crank–Nicholson scheme. Again we seek a variant that leads to a tridiagonal system. Such

a scheme is defined by $\beta_1^d = \beta_2^d = 0$ [fully explicit u equation (forward); this is necessary for a tridiagonal structure], and arbitrary β_3^d, β_4^d [i.e., w equation (backward) and vertically Crank–Nicholson]. One could seek a combination of the off-centering parameters[1] leading to a higher-order scheme. But divergence damping is not a physical process and therefore does not have to be treated very accurately. For simplicity we choose the values $\beta_3^d = \beta_4^d = 1$ in the following. This leads to the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = (1 - S_x^2)/(1 + S_z^2)$ and is therefore stable for $C_{\mathrm{div},x} < ½$ and arbitrary $C_{\mathrm{div},z}$.

4. Amplification factors for the 2D Euler equations

a. The three-stage Runge–Kutta scheme RK3WS

Now we combine the discretizations discussed so far. For the calculation of the amplification factors the matrices for advection in Eq. (21), sound in Eqs. (24) and (25), buoyancy in Eqs. (27) and (28), and divergence damping in Eq. (29) are combined to amplification matrices $\mathbf{Q}_s$ and $\mathbf{Q}_f$ and put into the operator formulation in Eq. (13) of the RK3WS scheme. Then the eigenvalues of $\mathbf{Q}_{\mathrm{RK3WS}}$ are calculated numerically by scanning the possible values for $k_x\Delta x \in [-\pi, \pi]$ with 601 equidistant points and for $k_z\Delta z \in [-\pi, \pi]$ with 101 equidistant points. These values are chosen to cover the current or soon to be used horizontal and vertical resolutions of weather prediction and climate models. Then the maximum amplification factor $\lambda_{\max}$ is determined as the largest absolute value from the $4 \times 601 \times 101$ eigenvalues $\lambda_i(k_x, k_z)$, $i = 1, \ldots, 4$ of the system. Those are calculated via the subroutine "zgeev" of the LAPACK package (Anderson et al. 1999).

The first question that arises is how to combine the fast processes. Intuitively a simple operator splitting (standard fractional step splitting) of the fast processes could be used, for example, in the sequence sound, divergence damping, and buoyancy:

$$\mathbf{Q}_f = \mathbf{Q}^{\mathrm{buoy}} \cdot \mathbf{Q}^{\mathrm{div}} \cdot \mathbf{Q}^{\mathrm{snd}}. \tag{30}$$

As we discussed in section 2, it is generally advantageous for the temporal order of the time-split system, if the fast processes are of second order in time. Operator splitting of noncommutable operators destroys this second-order convergence. On the other hand a simple "addition of tendencies" of two second-order discretizations (such as forward–backward or purely Crank–Nicholson, i.e., trapezoidal) is of second order. Indeed, it could be advantageous to use an addition of tendencies in the following form:

$$\mathbf{Q}_f = (\mathbf{1} + \mathbf{P}_1^{\mathrm{snd}} + \mathbf{P}_1^{\mathrm{div}} + \mathbf{P}_1^{\mathrm{buoy}})^{-1} \cdot (\mathbf{1} + \mathbf{P}_2^{\mathrm{snd}} + \mathbf{P}_2^{\mathrm{div}} + \mathbf{P}_2^{\mathrm{buoy}}). \tag{31}$$

Besides, this is slightly more efficient because the solution of tridiagonal systems has to be done only once in every small time step rather than being solved for each fast process separately. A comparison between these two methods is discussed later on (see Fig. 6). If not mentioned otherwise, the following analysis uses the addition of tendencies for the fast processes in Eq. (31).

For the beginning, we neglect the buoyancy terms and consider only advection with the fifth-order upwind scheme and sound terms. Figure 1 shows several coefficients for the divergence damping. A time-splitting ratio of $n_s = 6$ and a grid spacing ratio of $\Delta x/\Delta z = 10$ are chosen. Without divergence damping, the whole time-integration scheme is unstable, unless pure advection or pure sound propagation is considered (i.e., in the vicinity of the axes). Increasing $C_{\mathrm{div},x}$ stabilizes the scheme. But, otherwise, it reduces the maximum allowable small time step more and more. As for the sound waves, nearly horizontally propagating short waves become unstable first. Here $C_{\mathrm{div},x} = 0.1, \ldots, 0.15$ seems to be advisable, to stably integrate for $C_{\mathrm{snd},x} \approx 0.5, \ldots, 0.6$. For COSMO-DE a divergence damping coefficient of $\alpha_{\mathrm{div}} = 0.1(2800\ \mathrm{m})^2/(5\ \mathrm{s}) \approx 160\ 000\ \mathrm{m}^2\ \mathrm{s}^{-1}$ should be used. This leads to $\alpha_{\mathrm{div}}/c_s^2/\Delta t \approx 0.3$, which is larger than the value of 0.1 proposed in Wicker and Skamarock (2002).

Furthermore, the sound processes can be integrated without off-centering ($\beta_{2,4,6}^s = ½$); that is, a pure second-order Crank–Nicholson (trapezoidal) scheme can be used. Stronger off-centering has been tested, too, but it has almost no effect on $\lambda_{\max}$.

This holds with the existence of buoyancy terms, too, which are calculated in Figs. 2 and 3 with the generalized Crank–Nicholson scheme for a standard atmosphere $C_\beta = -0.24$ and a small time step of about 5 s; therefore $C_{\mathrm{buoy}} = 0.15$. The buoyancy terms, in contrast, need an off-centering of about $\beta^B = 0.7$, whereas a stronger off-centering does not increase stability. It can be seen further that RK3WS has a slight instability in the presence of buoyancy terms. Though $\lambda_{\max}$ less than about 1.0001–1.0003, which means an amplification of 0.01%–0.03% per time step, can be tolerated in limited-area models running at least in a weather forecast mode due to the limited time the most disturbances remain in the model domain.

Figure 4 shows $\lambda_{\max}$ for individual waves with a wave vector $(k_x\Delta x, k_z\Delta z)$ for the parameters $C_{\mathrm{adv}} = 1.0$,

[1] One can show (analogous to the generalized Crank–Nicholson scheme in the second section of appendix A) that sufficient conditions for stability are $C_{\mathrm{div},x} \le ½$, $½ \le \beta_4^d \le 1$, $\beta_4^d \le \beta_3^d \le 1$, and arbitrary $C_{\mathrm{div},z}$. This is quoted here without a proof for brevity.

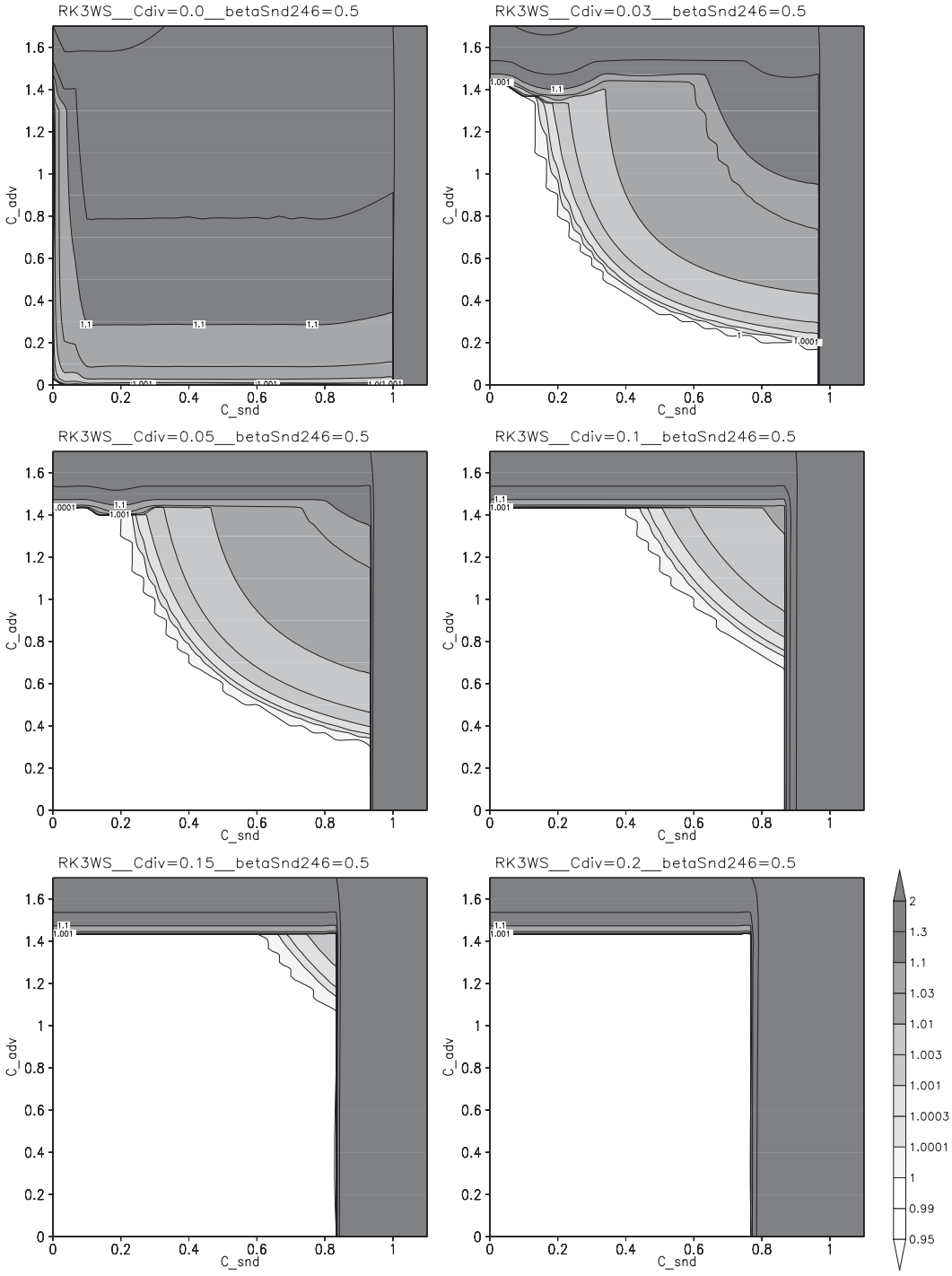

FIG. 1. Plot of $\lambda_{\max}$ as a function of $C_{\mathrm{snd},x}$ and C_{adv} for RK3WS with advection, sound, and divergence damping (without buoyancy) for (top left to bottom right) $C_{\mathrm{div},x} = 0$, 0.03, 0.05, 0.1, 0.15, and 0.2. The sound off-centering coefficient is $\beta^{s}_{2,4,6} = ½$ (i.e., trapezoidal). White color means stable range ($\lambda_{\max} \leq 1$). Isolines are at 1.0, 1.0001, 1.0003, 1.001, 1.003, and 1.01.

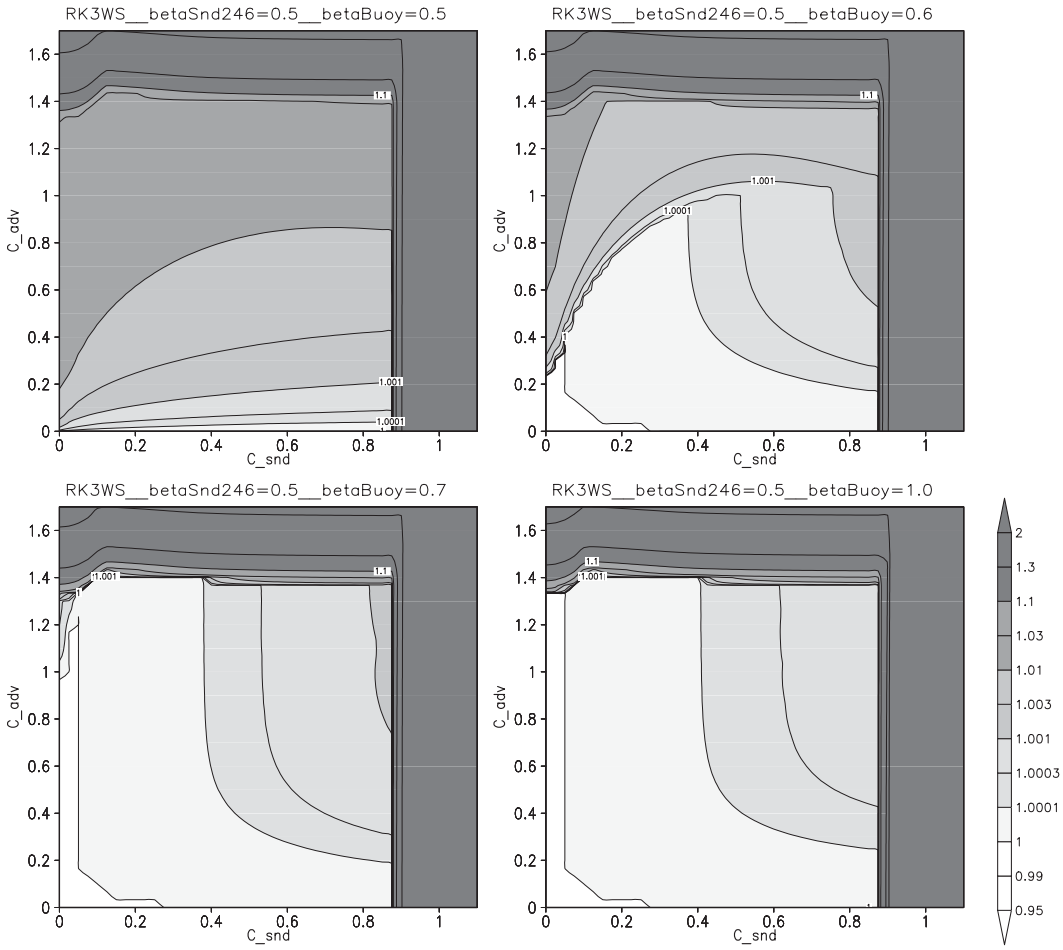

FIG. 2. As in Fig. 1, but with buoyancy. The $\lambda_{\max}$ for different off-centering for $\beta^B = 0.5$, 0.6, 0.7, and 1.0 and for $\beta^s_{2,4,6} = 0.5$ The time-integration scheme RK3WS, $C_{\mathrm{div},x} = 0.1$.

$C_{\mathrm{snd},x} = 0.6$, and again for a standard atmosphere and $\beta^B = 0.7$. Only positive values for k_x and k_z are shown because $\lambda_{\max}$ is symmetrical along the coordinate axes.[2] Without sound off-centering and without divergence damping, the instability of most of the waves is very strong. Both damping methods separately can remove the instability of nearly all the waves. But whereas sound off-centering alone retains mainly horizontally amplifying waves (both with short and long wavelengths), only divergence damping additionally removes instabilities of nearly horizontally propagating waves. A combination of sound off-centering and divergence damping does not improve stability, as mentioned above. What remains are weakly unstable vertically propagating waves because of the buoyancy terms.

Figure 5 shows $\lambda_{\max}$ for $C_{\mathrm{adv},x} = 1.1$ and $C_{\mathrm{snd},x} = 0.6$ in dependency on the off-centering for sound $\beta^s_{2,4,6}$ and buoyancy β^B for two different strengths of divergence damping $C_{\mathrm{div},x}$. It is interesting that the divergence damping has a stabilizing effect for sound waves in the sense that it allows smaller values for $\beta^s_{2,4,6}$ even below the stability limit ½ (but remind the reduction in C_{snd} stated above). For $C_{\mathrm{div},x} = 0.1$ again $\beta^B \approx 0.7$ is found as a reasonable value (i.e., only a very small amplification of waves).

We addressed above the question of several combinations of the single processes in the fast waves. Instead of using the "addition of tendencies" [Eq. (31)], different forms of operator splitting in the fast waves are inspected. The sequence of Eq. (30) and its cyclic transpositions $\mathbf{Q}_f = \mathbf{Q}^{\mathrm{snd}} \cdot \mathbf{Q}^{\mathrm{buoy}} \cdot \mathbf{Q}^{\mathrm{div}}$ and $\mathbf{Q}_f = \mathbf{Q}^{\mathrm{div}} \cdot \mathbf{Q}^{\mathrm{snd}} \cdot \mathbf{Q}^{\mathrm{buoy}}$ lead to completely unstable schemes. Whereas

[2] The symmetry for $(k_x, k_z) \rightarrow (-k_x, -k_z)$ follows from the fact that under this transformation the matrices for the single processes $\mathbf{Q}^{\mathrm{adv}}, \mathbf{P}^{\mathrm{snd}}_{1,2}, \mathbf{P}^{\mathrm{buoy}}_{1,2}, \mathbf{P}^{\mathrm{div}}_{1,2}$ invert to their complex conjugates and so does the total stability matrix $\mathbf{Q}_{\mathrm{RK3WS}}$. Consequently all its eigenvalues invert to complex conjugates, too. The second symmetry for $(k_x, k_z) \rightarrow (-k_x, k_z)$ is harder to prove: one could recognize it from the complete eigenvalue equation of the amplification matrix $\mathbf{Q}_{\mathrm{RK3WS}}$, but this is a very complicated expression.

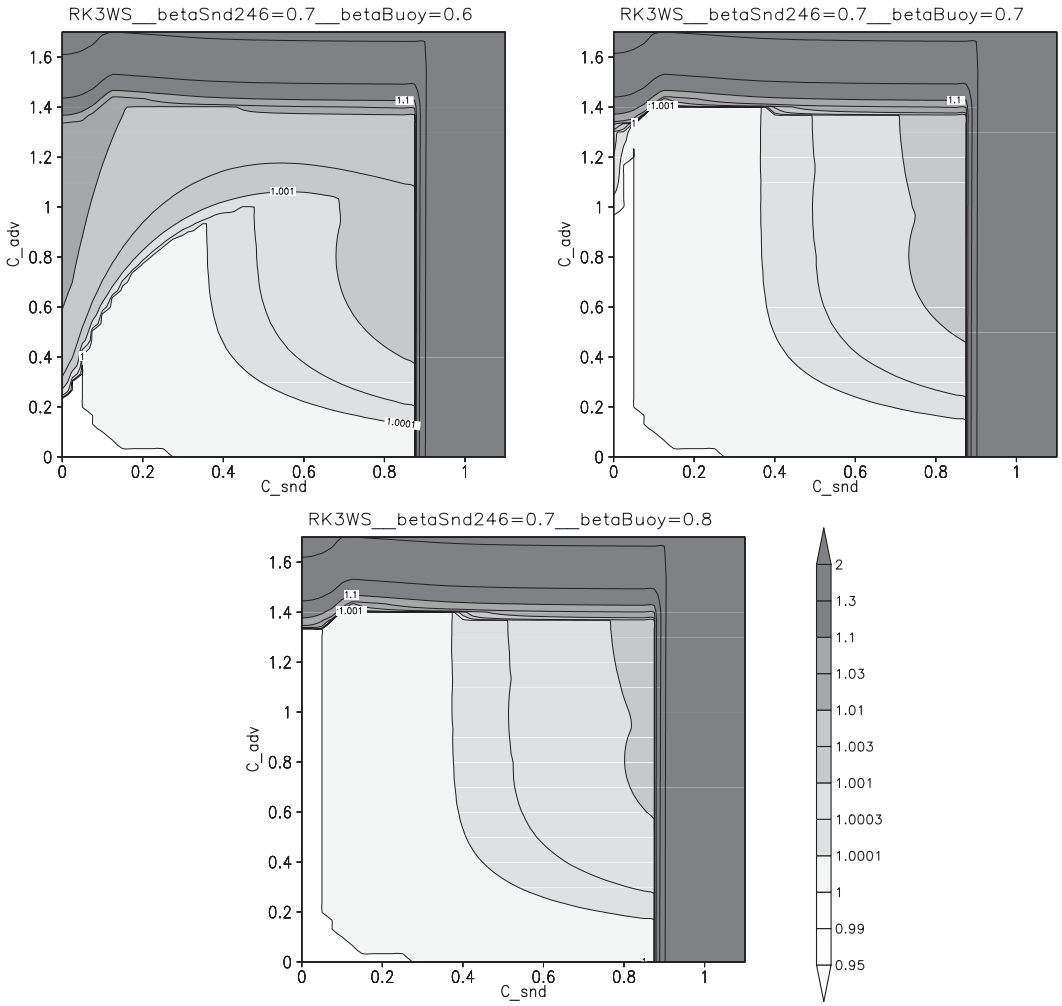

FIG. 3. As in Fig. 2, but for off-centering $\beta^s_{2,4,6} = 0.7$ and different values for $\beta^B = 0.6$, 0.7, and 0.8. The time-integration scheme RK3WS, $C_{\mathrm{div},x} = 0.1$.

$\mathbf{Q}_f = \mathbf{Q}^{\mathrm{div}} \cdot \mathbf{Q}^{\mathrm{buoy}} \cdot \mathbf{Q}^{\mathrm{snd}}$ (Fig. 6), and its cyclic transpositions on the one hand have a small range in $C_{\mathrm{snd},x}$ in which they are even more stable than the simple addition of tendencies, they are on the other hand strongly unstable for small $C_{\mathrm{snd},x}$. This extreme dependency on the sequence and the additional strong instability for weak flow cause operator splitting not to be a recommended option.

This can also be seen in Fig. 7, where $\lambda_{\max}$ in dependence on C_{buoy} and C_β (again for $C_{\mathrm{adv},x} = 1.1$, $C_{\mathrm{snd},x} = 0.6$, $\beta^s_{2,4,6} = 0.5$, and $\beta^B = 0.7$) for different $C_{\mathrm{div},x}$ for operator splitting in the sequence "sound–buoyancy–divergence damping" (right) is shown. Despite the fact that the "operator splitting" variant develops some stable region for larger C_{buoy}, it does not seem to be better than the addition of tendencies variant (left column in Fig. 7) for a time step of about $\Delta t = 5$ s and therefore $C_{\mathrm{buoy}} \approx 0.15$ and $C_\beta \approx -0.24$. Growing stability of the stratification (increasing C_β) leads to a slight reduction of the numerical stability. It is further remarkable that small values for the small time step Δt become more and more unstable if the divergence damping is too weak. This effect is very strong for the operator-splitting variant of the fast processes.

It should be noted that the correct onset of the physical instability for $C_\beta < -R/c_v \approx -0.4$ (i.e., unstable stratification) is produced by the stability analysis, too. This can serve as a test for correct programming of the analysis tool. This test is not entirely trivial, as the buoyancy terms alone do not produce this instability, but only the combination of sound and buoyancy for small wavelengths (i.e., short gravity waves).

The dependency of the stability on the grid spacing ratio has been inspected, too. These experiments have shown that the choice of a grid spacing ratio of $\Delta x/\Delta z = 10$ is reasonable. Smaller ratios up to 1, which are even larger than what is used in most applications, do not

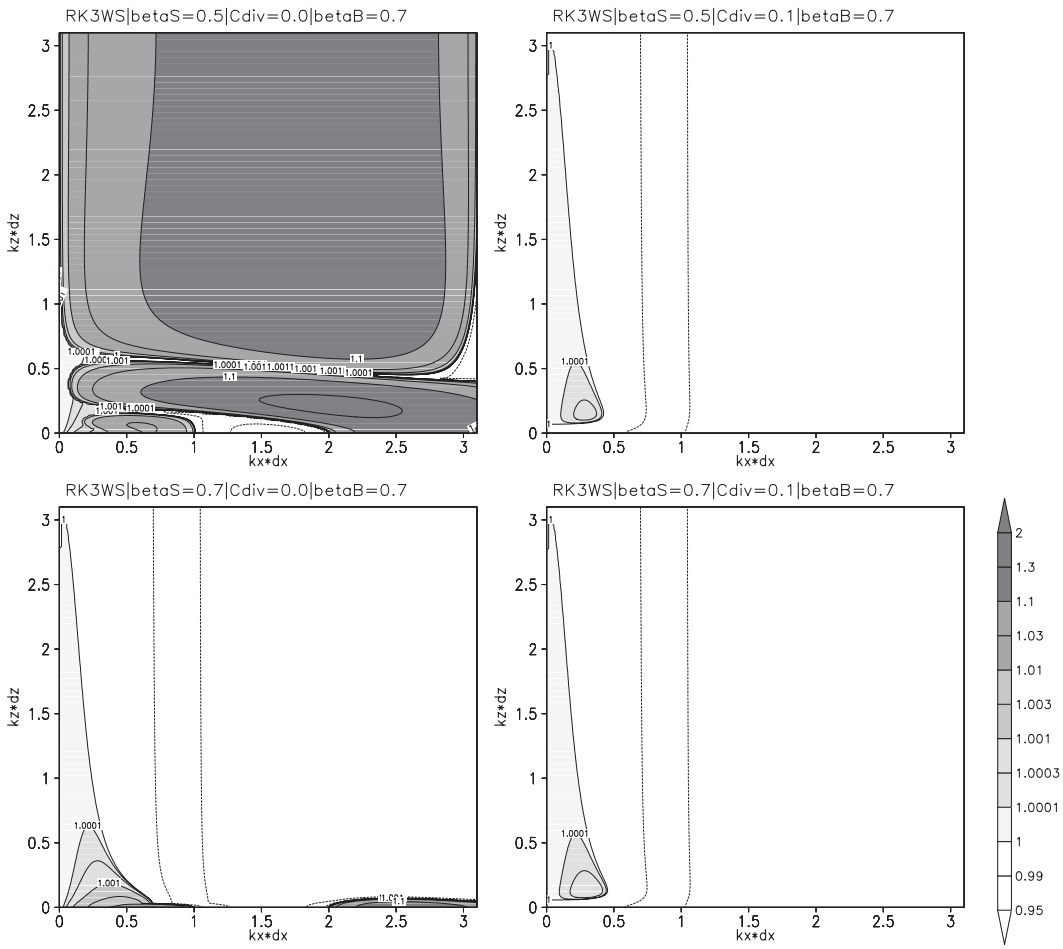

FIG. 4. Plot of $\lambda_{\max}$ for RK3WS as a function of $k_x\Delta x$ and $k_z\Delta z$ for $C_{\mathrm{adv}} = 1.0$, $C_{\mathrm{snd},x} = 0.6$ and for a standard atmosphere. Results for (left) $C_{\mathrm{div},x} = 0$ and (right) $C_{\mathrm{div},x} = 0.1$ and with (top) $\beta^s{}_{2,4,6} = ½$ or (bottom) $\beta^s{}_{2,4,6} = 0.7$.

further reduce the stable range. In contrast, larger ratios, which normally occur near the bottom boundary, even increase the stability of the time-splitting scheme.

Furthermore, it has been found that an increase of n_s leads to an increase of stability in the sense that the stable range tends toward a rectangle similar to the bottom right in Fig. 1. An explanation could be that for increasing $n_s \to \infty$ one expects a decoupling of slow and fast processes: the fast process acts as a noise on the slow process, whereas the slow process becomes more and more unimportant on the short time scale. The RK3WS scheme obviously supports this decoupling by an improvement of the stability properties. It is interesting that this decoupling property for RK3WS is indicated by the temporal Taylor expansion in Eq. (14), too (as seen above it becomes second order for $n_s \to \infty$).

Finally, the influence of additional artificial diffusion is studied. The fifth-order upwind advection operator [leading to Eq. (22)] contains an inherent diffusion proportional to C^4_{adv}. However, this only acts into the flow direction; hence, it does not prevent shear instabilities acting perpendicular to the flow. Therefore, a small amount of smoothing is recommended. This is done by a fourth-order artificial diffusion $-K_{\mathrm{smo}}\partial^4\phi/\partial x^4$ added on the rhs of the 2D Euler Eqs. (15)–(18) for every single prognostic variable and is applied outside of the RK scheme with a Marchuk operator splitting (i.e., after a large time step ΔT). The 4×4 amplification matrix $\mathbf{Q}^{\mathrm{smo}}$ is a multiple of the identity matrix with the diagonal elements $1 - C_{\mathrm{smo}}[2\cos(2K_x) - 8\cos(K_x) + 6]$, $K_x = k_x\Delta x$, and a dimensionless number $C_{\mathrm{smo}} = K_{\mathrm{smo}}\Delta T/\Delta x^4$. This scheme is stable and nonoscillating for $C_{\mathrm{smo}} \leq 1/16 = 0.0625$ (Durran 1998).

Figure 8 shows a small additional stabilization when compared with Fig. 2. The slight reduction in C_{adv} (Fig. 2) is caused by short waves and can therefore be enhanced by artificial diffusion up to the stability limit $C_{\mathrm{adv}} = 1.435$. The very weak instability for small C_{snd} or small C_{adv} is

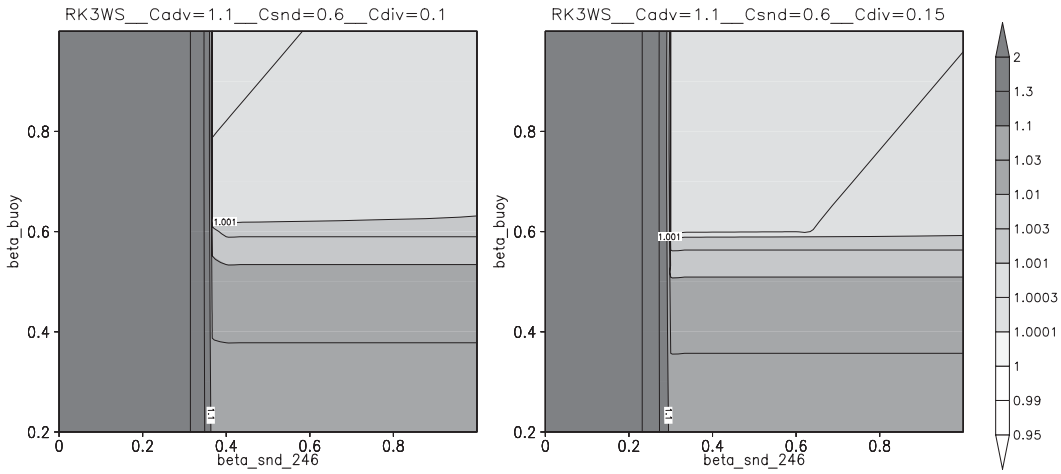

FIG. 5. Plot of $\lambda_{\max}$ as a function of $\beta^S_{2,4,6}$ and β^B for $C_{\mathrm{adv},x} = 1.1$ and $C_{\mathrm{snd},x} = 0.6$. (left) $C_{\mathrm{div},x} = 0.1$ and (right) $C_{\mathrm{div},x} = 0.15$.

also induced by short waves and can be removed by diffusion. The weak instabilities in the interior of the stability diagram are induced also by longer waves and therefore can hardly be removed by additional diffusion.

b. Third-order TVD Runge–Kutta

Shu (1988) found that the three-stage third-order RK scheme

$$\begin{aligned} q^* &= q^n + \Delta t f(q^n), \\ q^{**} &= \frac{3}{4}q^n + \frac{1}{4}q^* + \frac{1}{4}\Delta t f(q^*), \\ q^{n+1} &= \frac{1}{3}q^n + \frac{2}{3}q^{**} + \frac{2}{3}\Delta t f(q^{**}) \end{aligned} \tag{32}$$

has extraordinary nonlinear stability properties: if an Euler-forward discretization for f, which is strongly stable (e.g., TVD) for a time step ΔT is inserted into this RK scheme, then the whole scheme is strongly stable for the same time step. Therefore, he denoted this scheme TVD-RK3. While the fifth-order upwind advection operator [which leads to Eq. (22)] is not a stable EF scheme (it needs RK for stabilization), TVD-RK3 was nevertheless used as an alternative to RK3WS in Förstner and Doms (2004). The difference to the EF and RK3WS schemes discussed so far is the fact, that now the single RK steps in Eq. (32) do not start from the same state $\mathbf{q}^n$, but from different intermediate states. As mentioned in Ruuth and Spiteri (2004), there are several possibilities to formulate the same RK scheme—they denoted the TVD-RK3 as strong stability preserving [SSP](3, 3) RK scheme—which leads to different time-splitting formulations. Starting from the form provided in Shu (1988) and applying the same methodology as used in the sections before leads to the following operator:

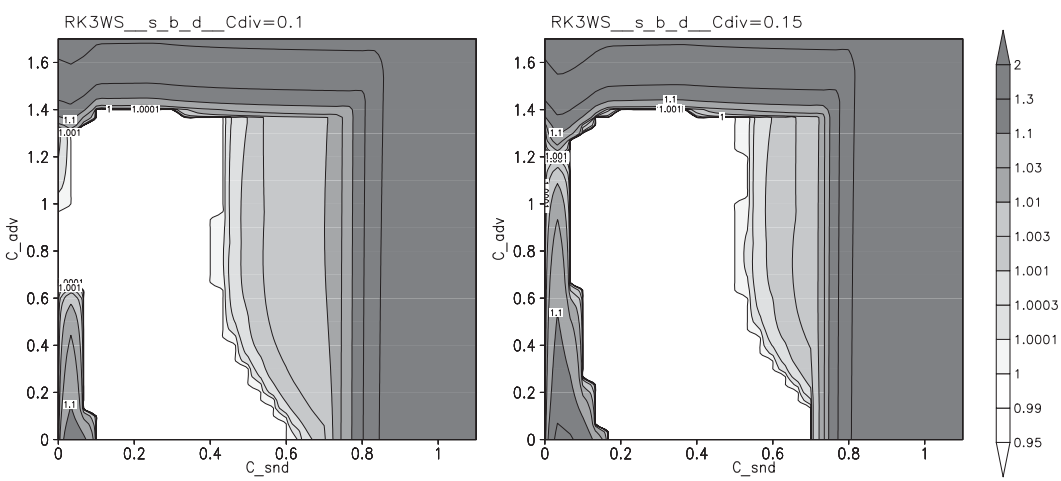

FIG. 6. Plot of $\lambda_{\max}$ for operator splitting in the fast processes in the sequence sound–buoyancy–divergence damping: (left) $C_{\mathrm{div},x} = 0.1$ and (right) 0.15.

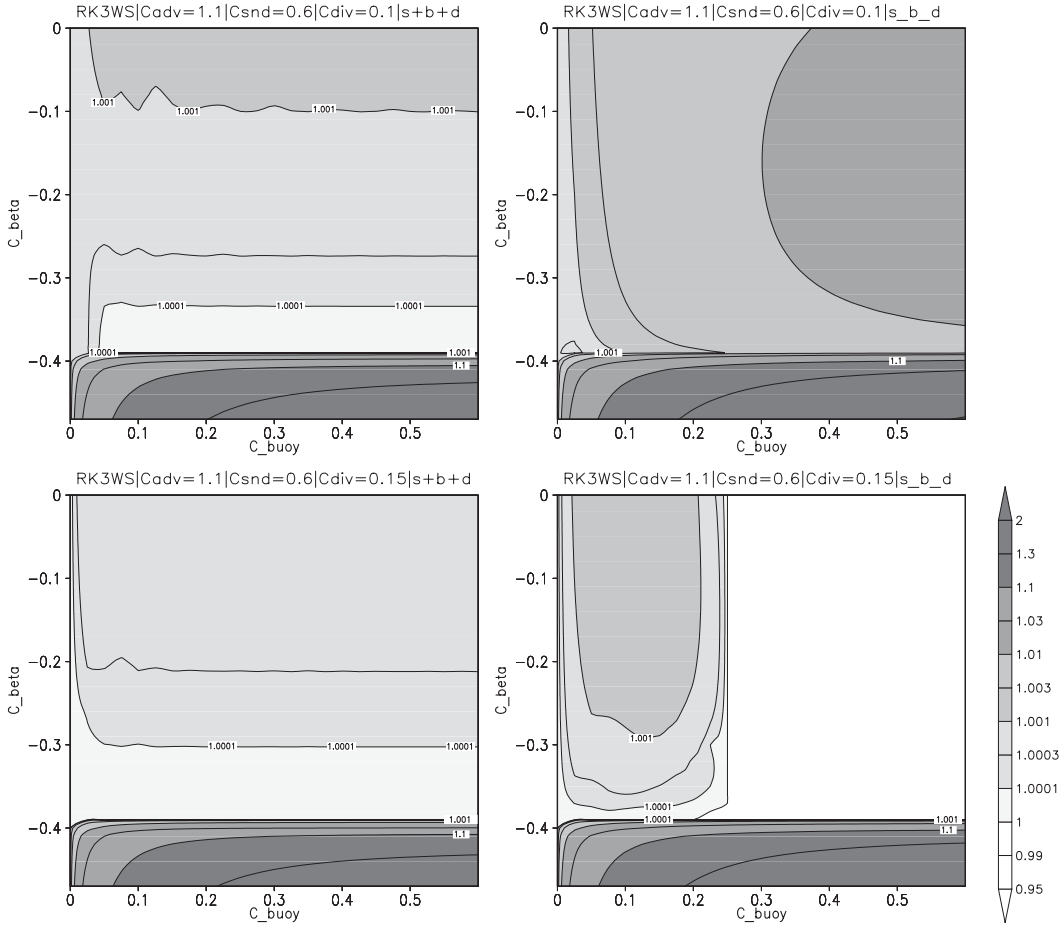

FIG. 7. Plot of $\lambda_{\max}$ as a function of C_{buoy} and C_β for $C_{\text{adv},x} = 1.1$ and $C_{\text{snd},x} = 0.6$ and different $C_{\text{div},x} = 0.1, 0.15$. Addition of (left) fast process tendencies or (right) operator splitting in the sequence sound–buoyancy–divergence damping.

$$
\begin{aligned}
Q_{1.\text{RK}} &:= Q_f^{n_s} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s-1} (Q_f)^i\right](Q_s - 1),\\
Q_{2.\text{RK}} &:= Q_f^{n_s/4}\left(\frac{3}{4}1 + \frac{1}{4}Q_{1.\text{RK}}\right)\\
&\quad + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/4-1} (Q_f)^i\right](Q_s - 1)Q_{1.\text{RK}},\\
Q_{\text{RK3TVD}} &:= Q_f^{2n_s/3}\left(\frac{1}{3}1 + \frac{2}{3}Q_{2.\text{RK}}\right)\\
&\quad + \frac{1}{n_s}\left[\sum_{i=0}^{2n_s/3-1} (Q_f)^i\right](Q_s - 1)Q_{2.\text{RK}}.
\end{aligned}
\tag{33}
$$

The computation amount per time step is approximately $3\tau_s + (23/12)n_s\tau_f$ and thus is only slightly greater than for the RK3WS scheme. The truncation error of this scheme is

$$
\begin{aligned}
Q_{\text{RK3TVD}} = 1 + \Delta T(P_s^{(1)} + P_f^{(1)}) + \frac{\Delta T^2}{2}\Bigg[&P_s^{(2)} + (P_s^{(1)})^2\\
&+ \left(\frac{85}{72} - \frac{1}{n_s}\right)P_f^{(1)}P_s^{(1)} + P_s^{(1)}P_f^{(1)}\\
&+ \left(\frac{85}{72} - \frac{1}{n_s}\right)(P_f^{(1)})^2 + \frac{1}{n_s}P_f^{(2)}\Bigg] + O(\Delta T^3).
\end{aligned}
\tag{34}
$$

Like RK3WS, RK3TVD is never of third order. It would be exactly of second order in time for $n_s = 72/13 \approx 5.5$ again under the condition that the slow process is an Euler-forward discretization [more precisely that $P_s^{(2)} = 0$] and the fast process is of second order in time. On the other hand, following the line of argument of section 4a, this indicates that RK3TVD does not support the decoupling of slow and fast processes for $n_s \to \infty$ and we would not expect an improvement in the stability properties for increasing n_s. Indeed, the stability analysis for

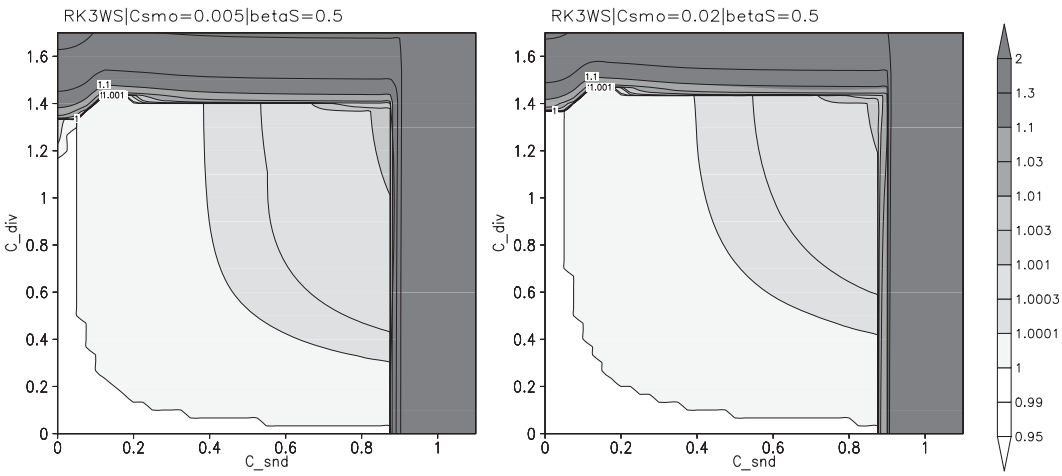

FIG. 8. As in Fig. 2, but $\lambda_{\max}$ for RK3WS ($\beta^B = 0.7$, $C_{\mathrm{div},x} = 0.1$) now with artificial diffusion with (left) $C_{\mathrm{smo}} = 0.005$ and (right) $C_{\mathrm{smo}} = 0.02$.

different $n_s = 12, 24$, and 48 shows, in contrast to RK3WS, that the stability decreases strongly with increasing time-splitting ratio.

An advantage of this scheme is the need of a slightly smaller divergence damping in combination with no off-centering for sound and buoyancy (Fig. 9, top left) for the price of a slightly reduced stable advection Courant number. But one should observe that because of the fractions of the time steps, only multiples of 12 can be used for n_s. This reduces the efficiency of the scheme, if used in meteorological applications. To circumvent this problem, one could use different small time steps (Förstner and Doms 2004). This is done here in the second RK step by using $\Delta t' = 3/4\Delta t$ in the calculation of Q_f and by using $n_s/3$ steps (instead of $n_s/4$) in the Q_f exponent and the sum, which allows $n_s = 6$. The computation amount per time step of this modified RK3TVD scheme is approximately $3\tau_s + 12/6\ n_s\tau_f$. Figure 9 (bottom right) shows that there is a bigger area of the lower left part in the stability diagram without spurious instability. But there is a drastic reduction of stability in the advection for larger $C_{\mathrm{snd},x}$, which cannot even be cured by stronger off-centering of $\beta^s_{2,4,6} = 0.7$. This probably caused a model abort of COSMO-DE during the severe storm Kyrill in January 2007 and led us to the decision to use RK3WS.

c. *The time-split "most simple" four-stage RK scheme*

Now we want to inspect a possible alternative to the three-stage RK schemes. The most simple time-split LC-RK4 scheme (as denoted in Baldauf 2008)—a fourth-order scheme for linear equations, but only second order for nonlinear problems—can be expressed as

$$Q_1 := Q_f^{n_s/4} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/4-1} (Q_f)^i\right](Q_s - 1),$$

$$Q_2 := Q_f^{n_s/3} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/3-1} (Q_f)^i\right](Q_s - 1)Q_1,$$

$$Q_3 := Q_f^{n_s/2} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s/2-1} (Q_f)^i\right](Q_s - 1)Q_2,$$

$$Q_{\mathrm{RK4MS}} := Q_f^{n_s} + \frac{1}{n_s}\left[\sum_{i=0}^{n_s-1} (Q_f)^i\right](Q_s - 1)Q_3.$$

It will be denoted as RK4MS in the following. The computation amount per time step is approximately $4\tau_s + (25/12)n_s\tau_f$. The truncation error for Q_{RK4MS} is up to the order ΔT^3 exactly as for Q_{RK3WS} [see Eq. (14)] and therefore leads to the same order conditions for the slow and fast processes; in particular, it becomes of second order for $n_s \to \infty$ and is never of third order. As for RK3WS, the stability increases for increasing n_s (not shown).

As derived in Baldauf (2008) the combination of a four-stage RK scheme with a centered difference advection scheme of fourth order with the dimensionless wavenumber

$$d(K) = -\frac{1}{6}\sin 2K + \frac{4}{3}\sin K$$

[used in the following instead of Eq. (22)] allows extraordinary large advection Courant numbers up to $C_{\mathrm{adv}} = 2.061$, which overcompensates the additional RK substep. Besides, this larger allowable ΔT could compensate for the

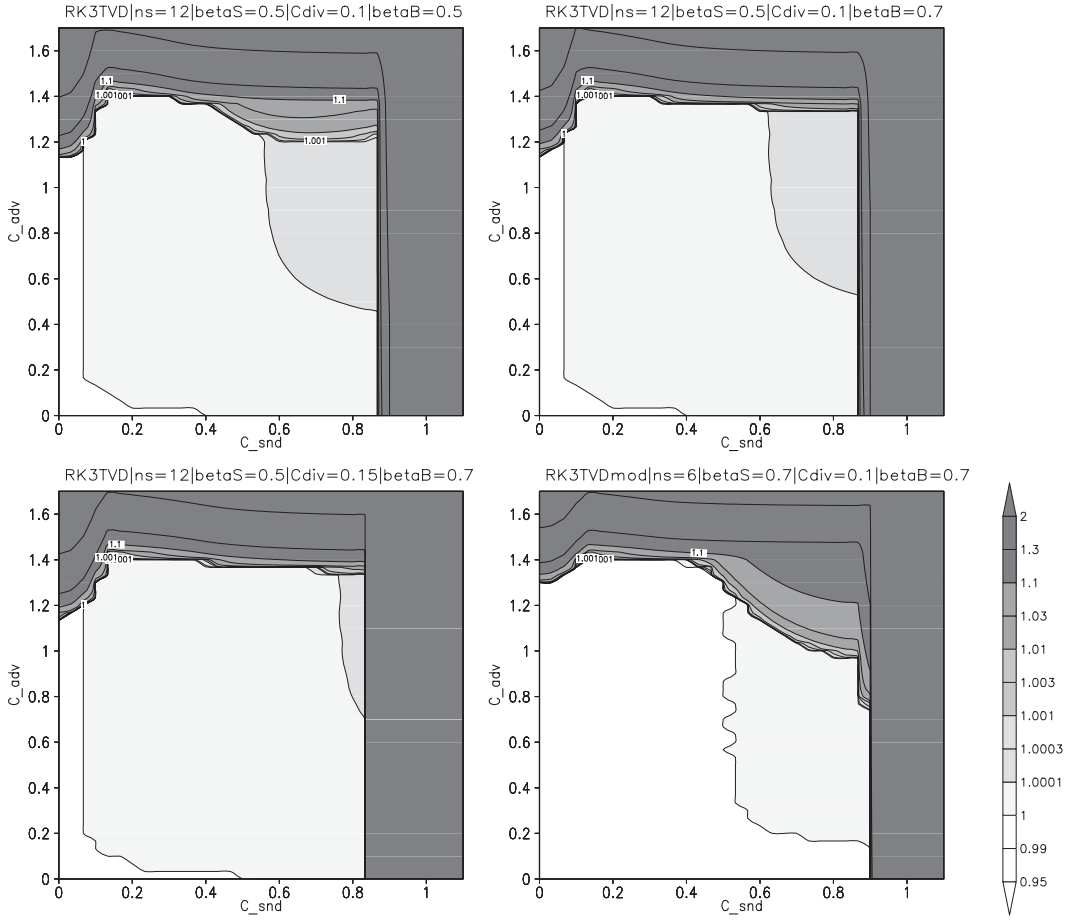

FIG. 9. Plot of $\lambda_{\max}$ for RK3TVD with $n_s = 12$, $\beta^s_{2,4,6} = 0.5$. (top) $C_{\mathrm{div},x} = 0.1$, (left) $\beta^B = 0.5$, and (right) $\beta^B = 0.7$. (bottom left) $C_{\mathrm{div},x} = 0.15$, $\beta^B = 0.5$. (bottom right) the modified RK3TVD for $n_s = 6$, $C_{\mathrm{div},x} = 0.1$, $\beta^s_{2,4,6} = 0.7$, and $\beta^B = 0.7$.

fact that this scheme can only use n_s as integer multiples of 12. Additionally a fourth-order advection scheme needs slightly less operations than a fifth-order scheme.

Generally a similar result is found as for RK3WS: the stability is quite independent from the off-centering of sound waves, so again $\beta^s_{2,4,6} = ½$ can be chosen. Furthermore, the buoyancy terms need some off-centering ($\beta^B = 0.7$).

The instabilities that occur (Fig. 10, top left) are generated by short waves and are due to the fact that a spatial centered difference scheme has no internal diffusion (in contrast to upwind biased schemes). Therefore, the fourth-order artificial diffusion $-K_{\mathrm{smo}}\partial^4\phi/\partial x^4$ as mentioned at the end of section 4a was applied. Figure 10 demonstrates that a weak diffusion with $C_{\mathrm{smo}} = 0.005$ delivers satisfying stability. For small C_{adv} or small C_{snd} the diffusion causes an analogous stabilization as for the RK3WS scheme in Fig. 8.

One should observe that due to the larger time-splitting ratio, smaller Δt occur. Applied to our COSMO-DE model with a horizontal grid spacing of 2.8 km, a $\Delta T \approx 35$ s could be achieved and with $n_s = 12$ and $\Delta t = 3$ s results. The latter leads to a sound Courant number of $C_{\mathrm{snd},x} = 0.35$. Therefore slightly larger amplification for the same $C_{\mathrm{snd},x}$ than for RK3WS can be tolerated. The dependence on the divergence damping is not very strong and satisfying results can be achieved with a smaller $C_{\mathrm{div},x} = 0.1$. Furthermore, for small advection Courant numbers there is no spurious instability.

5. Conclusions

A main goal of this work is to limit the parameter space for the occurring off-centering and artificial diffusion coefficients for the RK-based split-explicit schemes using forward–backward and generalized Crank–Nicholson schemes for the fast processes. This has been done by looking at the stability properties of single-process discretizations on the one hand and of a combined 2D nonhydrostatic, compressible Euler system on the other hand.

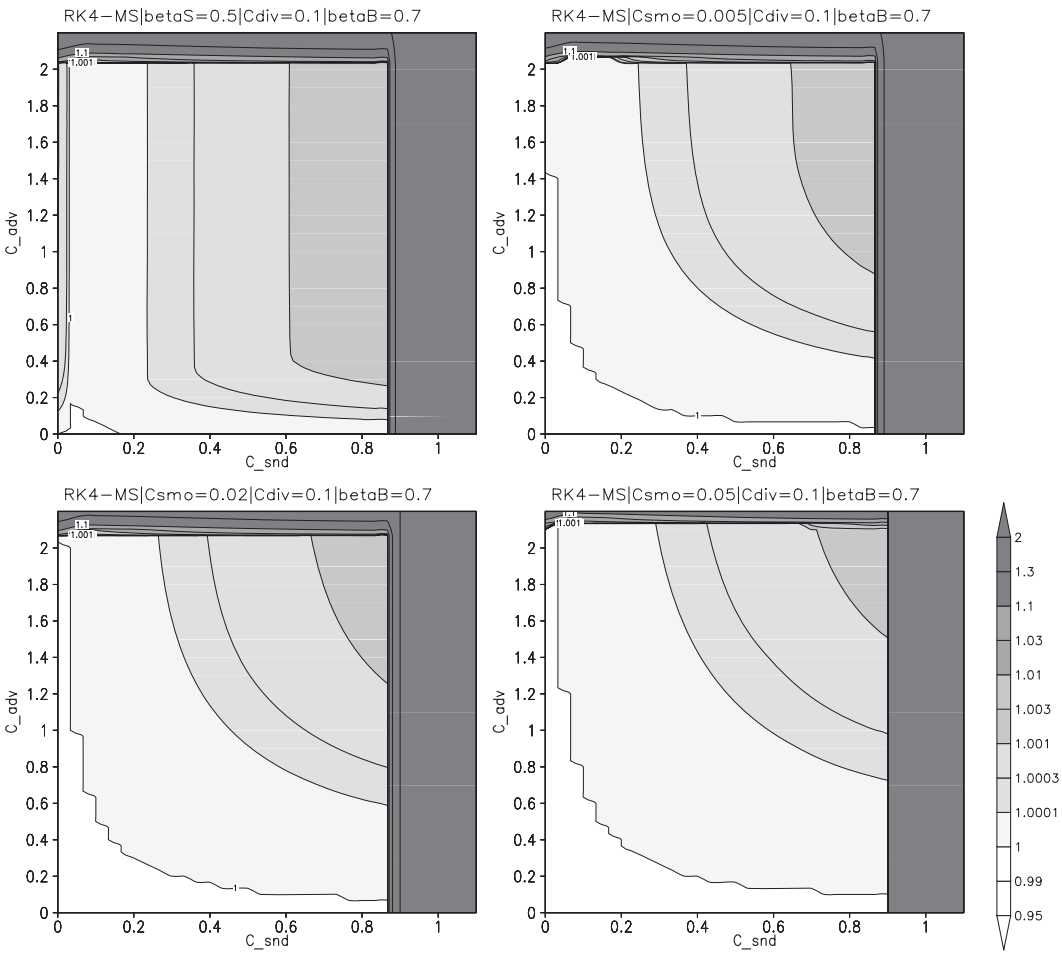

FIG. 10. Plot of $\lambda_{\max}$ for RK4MS with cd4 advection, $C_{\mathrm{div},x} = 0.1$, with different smoothing $C_{\mathrm{smo}} = 0$, 0.005, 0.02, and 0.05.

Operator splitting is quite incompatible with Runge–Kutta-based time splitting. This holds for multidimensional advection (Baldauf 2008) and was found here for the fast process, too. Instead, it is advantageous to use the "addition of tendencies" in the fast processes with no off-centering for sound $\beta^s_{2,4,6} = ½$ (a "pure" second-order Crank–Nicholson scheme) and an off-centered generalized Crank–Nicholson discretization for the buoyancy terms with $\beta^B_{1,2,3,4} = 0.7$. It should be mentioned again that no metric terms of a possible terrain-following coordinate are considered in this study. It is known from practical experiments that with metric terms a slightly stronger off-centering for the sound processes is advisable. This issue should be inspected in an extended study. The divergence damping should not be too small with $C_{\mathrm{div},x} = 0.1, \ldots, 0.15$.

In the RK3TVD scheme, the slight instability in the stable range is smaller than for the RK3WS scheme, but it has a reduction of efficiency due to time-step ratios n_s as integer multiples of 12 instead of 6. Any modifications to this scheme should be considered carefully because they can drastically reduce the stability for larger small time steps.

A further alternative to RK3WS with promising features is the four-stage, second-order RK scheme (RK4MS) in combination with a centered difference advection scheme of fourth order. It has quite similar stability properties as RK3WS with less instability for the smaller advection Courant numbers. It allows about a 45% bigger large time step, which overcompensates the use of an additional RK substep.

The formal order of temporal truncation error shows that it is generally difficult to achieve exactly second order in time even for the higher-order split RK schemes. The splitting error between fast and slow processes prohibits this. Therefore, the fact that RK4MS is only second order for nonlinear equations is not an obstacle.

A certain similarity was found between the temporal convergence order and stability for increasing n_s, namely that both convergence order and stability increase for

RK3WS and RK4MS, whereas both decrease for RK3TVD. A possible reason for that is that RK3WS and RK4MS support the decoupling of slow and fast processes for $n_s \to \infty$, whereas RK3TVD does not.

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APPENDIX A

Stability Statements for Fast Processes

a. Stability properties of the sound discretization

In this section, more detailed information is provided about the stability properties of the sound term discretization of section 3b. The case of a fully *explicit* discretization (i.e., all $\beta_1^s = \cdots = \beta_6^s = 0$) is unconditionally unstable. The eigenvalues of (23) are $1, 1, 1 \pm 2ia$, in which the abbreviation

$$a := \sqrt{S_x^2 + S_z^2}$$

will be useful in the following.

The *forward–backward integration*, which can be done in the two variants $\{\beta_1^s = \beta_2^s = 0, \beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = 1\}$ or $\{\beta_1^s = \beta_2^s = 1, \beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = 0\}$, results in the eigenvalues $1, 1, 1 - 2a^2 \pm 2\sqrt{a^4 - a^2}$. Therefore, both are stable and neutral for $a \le 1$ or $C_x^2 + C_z^2 \le 1$.

The *forward–backward scheme* in the horizontal in combination with a *vertically implicit* scheme ($\beta_1^s = 0$, $\beta_3^s = 1$, $\beta_5^s = 1$) leads to the four eigenvalues

$$\lambda_{1,2} = 1, 1, \quad \lambda_{3,4} = 1 - 2\frac{b \pm \sqrt{b^2 - a^2 c}}{c}, \qquad \text{(A1)}$$

with $b := S_x^2 + (\beta_2^s + \beta_4^s)S_z^2$ and $c := 1 + 4\,\beta_2^s\beta_4^s S_z^2$. Obviously, the thermodynamic constants c_p, c_v, R do not influence the numerical stability. Equation (A1) seems to be hard to analyze for arbitrary β_2^s or β_4^s. But in the special case $\beta_2^s = \beta_4^s =: \beta^s$ then $b^2 - a^2\,c = (S_x^2 - 1)(S_x^2 + S_z^2) - S_x^2 S_z^2[1 - 4\,\beta^s\,(1 - \beta^s)]$, which is negative or zero for $|S_x| \le 1$, $0 \le \beta^s \le 1$ and arbitrary S_z. Therefore, the square root in $\lambda_{3,4}$ is purely imaginary and the absolute square is

$$|\lambda_{3,4}|^2 = 1 - 4S_z^2 \frac{2\beta^s - 1}{1 + 4S_z^2(\beta^s)^2},$$

which is less than 1 for $\beta^s > ½$. This means that the scheme is stable for $|C_x| \le 1$, $½ \le \beta^s \le 1$, and arbitrary C_z. For $\beta^s > ½$ it is damping, whereas for a pure Crank–Nicholson variant (i.e., $\beta_2^s = \beta_4^s = ½$; this leads to $|\lambda_{3,4}| = 1$ for all S_z and $|S_x| \le 1$). Therefore, this special case is also neutral; the absolute values of all eigenvalues are 1.

It should be noted here that other variants that also use u^n besides u^{n+1} (e.g., $\beta_1^s = 0$, $\beta_2^s = 1$, $\beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = ½$ or $\beta_1^s = 0$, $\beta_2^s = \beta_3^s = \beta_4^s = \beta_5^s = \beta_6^s = ½$) are unconditionally unstable because that would mean losing the forward–backward idea in the horizontal.

A fully *implicit* scheme (all $\beta_1^s = \cdots = \beta_6^s = 1$) has eigenvalues $1, 1, (1 \pm 2ia)/(1 + 4a^2)$ and is unconditionally stable, yet damping. But it leads to a fully implicit system of equations (not tridiagonal) and is therefore not well suited for the split-explicit schemes considered in this paper.

b. Stability properties of the buoyancy discretization

This section provides derivations for the stability statements of section 3c, that is, the eigenvalue structure of $\mathbf{Q}^{\text{buoy}} := (\mathbf{1} + \mathbf{P}_1^{\text{buoy}})^{-1} \cdot (\mathbf{1} + \mathbf{P}_2^{\text{buoy}})$. Again a fully *explicit* scheme ($\beta_1^B = \beta_2^B = \beta_3^B = \beta_4^B = 0$) is unconditionally unstable (more precisely, it would be stable only for $C_g = 0$ or $C_{\text{buoy}} = 0$, which are both not realistic).

A *forward–backward* scheme with two variants $\{\beta_1^B = \beta_2^B = 0$ and $\beta_3^B = \beta_4^B = 1\}$ or $\{\beta_1^B = \beta_2^B = 1$ and $\beta_3^B = \beta_4^B = 0\}$ delivers the eigenvalue equation

$$(\lambda - 1)^2[\lambda^2 - (2 - C_{\text{buoy}}^2)\lambda + 1] = 0,$$

leading to $\lambda_{1,2} = 1$, $\lambda_{3,4} = 1 - C_{\text{buoy}}^2/2 \pm \sqrt{-C_{\text{buoy}}^2 + C_{\text{buoy}}^4/4}$. Therefore, if $0 < C_{\text{buoy}}^2 < 4$, then all eigenvalues $|\lambda_i| = 1$ and the scheme is stable and neutral if $C_{\text{buoy}} < 2$.

The stability analysis for arbitrary β_i^B seems to be quite complicated but in the case of a *generalized Crank–Nicholson scheme*, in which all β_i^B are equal $\beta_1^B = \beta_2^B = \beta_3^B = \beta_4^B = \beta^B$, the problem is solvable. The eigenvalues are

$$\lambda_{1,2} = 1, \quad \lambda_{3,4} = \frac{1 - a^2\beta^B(1 - \beta^B) \pm ia}{1 + a^2(\beta^B)^2}, \qquad \text{(A2)}$$

with $a := S_g\sqrt{C_\beta + c_p/c_v}$. From this it follows that for $\beta^B < ½$ the scheme is unconditionally unstable [a special case is the "pure explicit" discretization discussed above ($\beta^B = 0$)]. Here $\beta^B = ½$ leads to an unconditionally stable and neutral scheme. For $\beta^B > ½$ the scheme is also unconditionally stable, but damping. The proof for this is easily done: the absolute square of

$$|\lambda_{3,4}|^2 = \frac{[1 - a^2\beta^B(1-\beta^B)]^2 + a^2}{[1 + a^2(\beta^B)^2]^2}$$

is symmetric around $a = 0$ and has the simple derivative

$$\frac{\partial|\lambda_{3,4}|^2}{\partial a} = 2a\frac{1-2\beta^B}{[1+a^2(\beta^B)^2]^2} = \begin{cases} >0 & \text{for} \quad \beta^B < ½ \\ <0 & \text{for} \quad \beta^B > ½ \end{cases}.$$

Therefore, $|\lambda_{3,4}|^2$ is a monotonically falling function in a if $\beta^B > ½$. As a result of

$$|\lambda_{3,4}|^2(a=0) = 1$$

and

$$\lim_{a\to+\infty}|\lambda_{3,4}|^2 = \left(\frac{1-\beta^B}{\beta^B}\right)^2 \le 1 \quad \text{for} \quad \beta \ge \frac{1}{2},$$

there follows the stability statement.

APPENDIX B

Consequences of a Bretherton Transformation

In the theory of linear waves it is convenient to perform a transformation introduced by Bretherton (1966) to take into account the quasi-exponential decay of pressure and density. The Bretherton transformation (BT) is performed by insertion of $(u, w, T') = (\tilde{u}, \tilde{w}, \tilde{T})(\rho_0/\rho_s)^{-1/2}$ and $p' = \tilde{p}(\rho_0/\rho_s)^{1/2}$ into Eqs. (15)–(19) (Gassmann and Herzog 2007). Here ρ_s is a constant reference value and an inverse scale height can be defined by $\delta := -d\log(\rho_0/\rho_s)/dz$.

The resulting equations are generated from Eqs. (15)–(19) if all derivatives $\partial/\partial z$ are replaced by $\partial/\partial z \pm \delta/2$ (+ for u, w, or T', − for p'), ρ_0 is replaced by ρ_s, and p_0 by $\rho_s R T_0$. One can alternatively consider Eqs. (15)–(19) as a BT system, in which terms $\sim\delta$ are neglected.

A Fourier transformation $\tilde{\phi}(x, z, t) = \tilde{\phi}(k_x, k_z, \omega)e^{i(k_x x + k_z z - \omega t)}$ generates a linear equation system. The determinant of its coefficient matrix delivers the characteristic equation

$$\tilde{\omega}^4 + i\alpha_{\text{div}}\left(k^2 - \frac{\delta^2}{4} - ik_z\delta\right)\tilde{\omega}^3 - \left[c_s^2\left(k^2 + \frac{\delta^2}{4}\right) + \omega_a^2 - g\delta\right]\tilde{\omega}^2 - i\alpha_{\text{div}}k_x^2(\omega_a^2 - g\delta)\tilde{\omega} + k_x^2c_s^2N^2 = 0. \quad \text{(B1)}$$

Here $\tilde{\omega} := \omega - U_0k_x$ denotes the Doppler-shifted frequency. In the case $\delta \neq 0$, one can prove that $\omega_a^2 - g\delta = 0$ and Eq. (B1) is identical to Eq. (15) of Gassmann and Herzog (2007). Nevertheless, this term is kept here because for $\delta = 0$ (i.e., no BT) Eq. (B1) describes the analytic wave solution discussed in this work. Here, we want to discuss two unstable limiting cases for $\delta \neq 0$.

For rigid oscillations with $\mathbf{k} = 0$ we get the solutions $\tilde{\omega}_{3,4} = 0$ for gravity waves and

$$\tilde{\omega}_{1,2} = \frac{1}{8}i\delta^2\alpha_{\text{div}} \pm \sqrt{\omega_a^2 - \delta g - \frac{1}{64}\delta^4\alpha_{\text{div}}^2 + \frac{1}{4}\delta^2c_s^2} \quad \text{(B2)}$$

for sound waves. The radicand is positive for usual stratifications and therefore the square root is real. The positive imaginary part in the first term shows that extremely long sound waves are unstable for a BT system that includes the artificial divergence damping.

The same holds for short gravity waves ($k \equiv |\mathbf{k}| \to \infty$) and for small α_{div} with the dispersion relation

$$\tilde{\omega}_{3,4} = \pm N\frac{k_x}{k} + \frac{i}{2}\frac{k_x^2}{k^2}\frac{N^2 - \omega_a^2 + g\delta}{c_s^2}\alpha_{\text{div}} + O\left(\frac{1}{k^2}, \frac{\alpha_{\text{div}}}{k}, \alpha_{\text{div}}^2\right). \quad \text{(B3)}$$

For $\delta = 0$ (i.e., the system inspected in this paper) the second term on the rhs has a negative sign and therefore has a (very weak) damping effect on gravity waves. And of course a BT system without divergence damping ($\alpha_{\text{div}} = 0$) is stable, too. But artificial divergence damping in combination with a BT shows a slight instability even in the analytic case. Therefore BT was not used here as a starting point to investigate numerical stability of split-explicit schemes.

REFERENCES

Anderson, E., and Coauthors, 1999: LAPACK users' guide. 3rd ed. Society for Industrial and Applied Mathematics, Philadelphia, PA, 407 pp.

Baldauf, M., 2002: Integration by time-splitting in mesoscale models. *COSMO Newsl.,* **2,** 101–104.

——, 2008: Stability analysis for linear discretisations of the advection equation with Runge–Kutta time integration. *J. Comput. Phys.,* **227,** 6638–6659.

Bretherton, F. P., 1966: The propagation of group of internal gravity waves in a shear flow. *Quart. J. Roy. Meteor. Soc.,* **92,** 466–480.

Doms, G., and U. Schättler, 2002: A description of the nonhydrostatic regional model LM. Part I: Dynamics and numerics. LM_F90 2.18, Deutscher Wetterdienst, Offenbach, Germany, 140 pp.

Durran, D. R., 1998: *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*. Springer, 482 pp.

Förstner, J., and G. Doms, 2004: Runge–Kutta time integration and high-order spatial discretization of advection—A new dynamical core for the LMK. *COSMO Newsl.,* **4,** 168–176.

Gassmann, A., and H.-J. Herzog, 2007: A consistent time-split numerical scheme applied to the nonhydrostatic compressible equations. *Mon. Wea. Rev.,* **135,** 20–36.

Jebens, S., O. Knoth, and R. Weiner, 2009: Explicit two-step peer methods for the compressible Euler equations. *Mon. Wea. Rev.,* **137,** 2380–2392.

Klemp, J. B., and R. B. Wilhelmson, 1978: The simulation of three-dimensional convective storm dynamics. *J. Atmos. Sci.,* **35,** 1070–1096.

——, W. C. Skamarock, and J. Dhudia, 2007: Conservative split-explicit time integration methods for the compressible nonhydrostatic equations. *Mon. Wea. Rev.,* **135,** 2897–2913.

LeVeque, R. J., and J. Oliger, 1983: Numerical methods based on additive splittings for hyperbolic partial differential equations. *Math. Comput.,* **40,** 469–497.

Mesinger, F., 1977: Forward-backward scheme, and its use in a limited area model. *Beitr. Phys. Atmos.,* **50,** 200–210.

Purser, R. J., and L. M. Leslie, 1991: Reducing the error in a time-split finite difference scheme using an incremental technique. *Mon. Wea. Rev.,* **119,** 578–585.

Ruuth, S. J., and R. J. Spiteri, 2004: High-order strong-stability-preserving Runge–Kutta methods with downwind-biased spatial discretizations. *SIAM J. Numer. Anal.,* **42,** 974–996.

Shu, C.-W., 1988: Total-variation diminishing time discretizations. *SIAM J. Sci. Stat. Comput.,* **9,** 1073–1084.

Skamarock, W. C., and J. B. Klemp, 1992: The stability of time-split numerical methods for the hydrostatic and the nonhydrostatic elastic equations. *Mon. Wea. Rev.,* **120,** 2109–2127.

——, and ——, 2008: A time-split nonhydrostatic atmospheric model for weather research and forecasting applications. *J. Comput. Phys.,* **227,** 3465–3485.

Steppeler, J., G. Doms, U. Schättler, H. W. Bitzer, A. Gassmann, U. Damrath, and G. Gregoric, 2003: Meso-gamma scale forecasts using the nonhydrostatic model LM. *Meteor. Atmos. Phys.,* **82,** 75–96.

Viner, K. C., and C. C. Epifanio, 2008: An analysis of Klemp–Wilhelmson schemes as applied to large-scale wave modes. *Mon. Wea. Rev.,* **136,** 4730–4745.

Wicker, L. J., and W. C. Skamarock, 1998: A time-splitting scheme for the elastic equations incorporating second-order Runge–Kutta time differencing. *Mon. Wea. Rev.,* **126,** 1992–1999.

——, and ——, 2002: Time splitting methods for elastic models using forward time schemes. *Mon. Wea. Rev.,* **130,** 2088–2097.